

Commodity Prices and Sovereign Risk^{*}

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Abstract

Swings in global commodity prices affect sovereign risk unevenly across countries. Using a shift-share strategy over 41 commodities, we show that import-price shocks raise sovereign spreads and reduce borrowing, while export-price shocks have weaker effects and tend to lower spreads. To explain this asymmetry, we develop a sovereign default model in which a common global factor drives prices. A surge in global prices affects risk through a price-index channel, which raises consumption costs, and a terms-of-trade income channel. Calibrating the model to 37 emerging economies, we show that trade structure predicts whether price booms widen or compress sovereign spreads.

JEL-classification: F34, F41, E32, E44, F62.

Key words: Sovereign Default, Commodity Prices, Terms-of-Trade, External Borrowing, Global Shocks, Spreads.

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1 Introduction

Sovereign defaults often occur in waves that coincide with global commodity price cycles. Yet, the same price movement can be a tailwind for some economies and a headwind for others. Commodity exporters benefit from price booms through improved terms of trade, higher real income, and lower borrowing costs, whereas import-dependent economies face precisely the opposite: higher global prices compress consumption, erode the external balance, and elevate default risk.

This paper argues that the direction of the sovereign risk response to global commodity price movements is determined by a single observable feature of each economy: whether it is a net exporter or importer of the goods driving global prices. Since increases in export prices raise real income and strengthen repayment incentives, while increases in import prices act as negative income shocks that compress consumption and elevate default risk, the effect of global price movements on aggregate consumption is governed by a country's trade structure.

We establish this result in three steps. First, using a shift-share identification strategy that exploits cross-country heterogeneity in trade baskets, we provide causal evidence of a sharp asymmetry: import price shocks raise sovereign spreads by 70–80 basis points per standard deviation and erode the current account, while export price shocks weakly compress spreads and improve the external balance. These results align with the view that terms-of-trade fluctuations act as real income shocks, with the resulting asymmetry in spreads reflecting heterogeneity in trade structure: export price booms compress spreads for net exporters, while for countries that both export and consume the same commodities, the income and price effects partially offset.

Second, we develop a sovereign default model that formalizes this mechanism. The model introduces *global price shocks* as the central source of risk and accounts for cross-country heterogeneity through two observable transmission channels. In contrast to income-based models such as [Arellano \(2008\)](#), where shocks affect output directly, global price movements here operate through two channels: (i) a *price-index channel*, which captures how changes in world prices alter the cost of the domestic consumption basket, and (ii) a *terms-of-trade income channel*, which captures how the same price movements affect real income through trade. The price-index channel reflects a cost-of-living effect: increases in the prices of highly traded goods raise the cost of consumption and reduce real income. The terms-of-trade channel reflects an income effect: for countries that export goods with high exposure to the global factor, global upturns increase export revenues, while for net importers the same shocks deteriorate the trade balance. Crucially, the interaction of the two channels differs across countries. In import-dependent economies, both channels operate in the same direction: higher global prices simultaneously raise the cost of the consumption basket and worsen the trade balance, reinforcing the decline in real income and amplifying default risk. In contrast, in exporter economies the two channels work in opposite directions: the

positive income effect from improved terms of trade partly offsets the adverse cost-of-living effect, dampening the overall response of consumption and sovereign risk. The net effect of these forces, which we refer to as the *consumption wedge*, determines whether a global price increase is expansionary or contractionary and therefore governs how global shocks transmit to sovereign risk. This asymmetry provides a structural interpretation of the empirical findings: import price shocks generate large and precisely estimated effects because the two channels reinforce each other, whereas export price shocks have weaker average effects due to the partial offset between income and price-index channels. The model embeds these mechanisms within a standard dynamic default framework with risk-neutral lenders and endogenous borrowing limits. Countries differ only in their trade exposure and consumption structure, so that the composition of trade alone determines which side of the wedge they fall on, while an endogenous leverage term amplifies the strength of the transmission.

Third, we calibrate the model for each of the 37 emerging economies in our sample, computing country-specific exposure parameters from international trade and production data. The cross-country exercise delivers a clean empirical test with no free parameters: the sign of spread cyclicity flips at the theoretical threshold: in 20 countries sovereign spreads fall when global commodity prices rise (exporter-type), and in 17 countries spreads widen (importer-type). In addition, the predicted sign matches EMBI data for the vast majority of the sample, and historical default episodes; Russia 1998, Ecuador 1999 and 2008, Argentina 2001, Venezuela 2014 and 2017 (exporter-type); Lebanon 2020, Sri Lanka 2022 (importer-type); align with the timing predicted by the model, suggesting that exporter defaults cluster at troughs of the global commodity factor and importer defaults at peaks. Leverage endogenously amplifies the transmission: economies with smaller precautionary buffers display stronger sensitivity of default risk and borrowing costs to global factors.

Together, these results imply that trade structure is a first-order determinant of sovereign resilience to global commodity cycles. Indeed, the effect of a global price shock on borrowing conditions is determined by a country’s trade position: shocks that relax constraints for exporters tighten them for importers of the same goods, independently of institutional characteristics, policy choices, or debt levels. Recognizing this asymmetry matters for sovereign risk assessment, for the design of international support mechanisms, and for understanding why commodity booms and busts generate waves of default that look puzzlingly heterogeneous in the cross-section.

Related Literature. This paper connects three active literatures: (i) *modern shift-share identification strategies for isolating exposure to common shocks*, (ii) *the macroeconomic consequences of terms-of-trade and commodity-price shocks in small open economies* and (iii) *the quantitative sovereign default tradition with endogenous spreads*.

We build on the shift-share designs of [Adao et al. \(2019\)](#), [Goldsmith-Pinkham et al. \(2020\)](#), and [Borusyak et al. \(2022\)](#) to study heterogeneous exposure to common shocks. Our

key contribution is to construct separate instruments for import and export exposure, which allows us to identify how the effect of global price shocks on sovereign spreads depends on trade structure. Relative to existing applications of these designs to commodity prices (e.g., [Drechsel & Tenreyro 2018](#)), we show that a common global shock induces a sign reversal in sovereign risk between net exporters and net importers. To our knowledge, this paper is the first to apply shift-share methods to sovereign spreads and external borrowing while preserving this import-export decomposition.

Terms-of-trade movements are a central driver of business-cycle fluctuations in small open economies ([Mendoza 1995](#), [Schmitt-Grohé & Uribe 2018](#), [Kose 2002](#), [Fernández et al. 2018](#)). Commodity price declines have been linked to sudden stops ([Caballero & Panageas 2008](#), [Calvo et al. 2008](#)), output volatility ([Drechsel & Tenreyro 2018](#)), and asymmetric responses of output and investment ([Ferroni et al. 2025](#)). These movements are largely global, driven by a small number of common factors ([Pindyck & Rotemberg 1990](#), [Byrne et al. 2013](#), [Kilian 2009](#), [Kilian & Murphy 2014](#), [Känzig 2021](#), [Baumeister & Hamilton 2019](#)), and closely associated with sovereign default waves over the past two centuries ([Reinhart et al. 2016](#)). We use principal components of international prices to capture these global shocks and show that their effects on sovereign risk depend systematically on trade structure, generating a sign reversal between exporters and importers. While existing work has focused predominantly on commodity exporters, for whom higher prices relax financing constraints, [Di Pace et al. \(2025\)](#) show that import- and export-price shocks have asymmetric macroeconomic effects. We build on this insight by formalizing this asymmetry in a sovereign risk setting and by identifying two transmission mechanisms, a price-index channel and a terms-of-trade income channel, whose relative strength determines whether global price increases are expansionary or contractionary.

Finally, we extend the canonical sovereign default framework of [Eaton & Gersovitz \(1981\)](#), [Aguiar & Gopinath \(2006\)](#), [Arellano \(2008\)](#), [Aguiar & Amador \(2014\)](#) and its standard quantitative extensions to long-maturity debt ([Hatchondo & Martinez 2009](#), [Chatterjee & Eyigungor 2012](#)), renegotiation ([Yue 2010](#)), production and external adjustment ([Mendoza & Yue 2012](#)), the joint determination of default and devaluation ([Na et al. 2018](#)), and financial-sector spillovers ([Bocola 2016](#), [Corsetti et al. 2013](#)). Our key departure is to replace exogenous income shocks with observable global price factors that enter both the consumption price index and trade income, with country-specific exposure determined directly by trade weights. A large literature links commodity prices to sovereign risk ([Min 1998](#), [Cuadra & Sapriza 2006](#), [Hilscher & Nosbusch 2010](#), [Roch 2019](#), [Hatchondo et al. 2007](#), [Shousha 2016](#), [Drechsel & Tenreyro 2018](#)) and emphasizes the role of global financial conditions in driving spreads ([González-Rozada & Yeyati 2008](#), [Longstaff et al. 2011](#), [Rey 2015](#), [Miranda-Agrippino & Rey 2020](#)), but focuses primarily on exporters. We show that the importer case is qualitatively different: global price increases can raise risk premia for net importers even holding fundamentals fixed, because the price-index channel dominates the income channel. Our framework delivers three contributions. First, we provide a structural decomposition

of default incentives into a price-index channel and a terms-of-trade income channel, which explains why spread cyclicity reverses sign between exporters and importers. Second, we discipline the quantitative analysis directly from data, as cross-country heterogeneity is pinned down by observed trade exposures rather than free parameters. Third, we generate the sign reversal in sovereign spreads as an equilibrium outcome. We solve the model for 37 emerging economies with country-specific calibrations, validate the predicted sign of spread cyclicity against EMBI data and historical default episodes, and show that cross-country variation in sovereign risk maps directly into the relative strength of the two channels.

The paper proceeds as follows. Section 2 develops the identification strategy and presents the empirical evidence, documenting the asymmetric effects of import and export price shocks on sovereign spreads, external borrowing, and the current account. Section 3 introduces the theoretical framework, formalizing the two channels through which global commodity prices affect sovereign risk and establishing conditions under which their relative strength determines the cyclicity of spreads. Section 4 brings the model to the data, calibrating it for each of the 37 countries in our sample and validating its cross-country predictions against EMBI spread data and historical default episodes; the full set of country-level results is collected in Appendix F. Section 5 concludes.

2 Empirical Evidence

2.1 Empirical Strategy

Our objective is to evaluate the effect of terms-of-trade movements on default risk and sovereign borrowing. We estimate regressions of the form:

$$\Delta y_{i,t} = \alpha_i + \beta_t + \gamma \Delta P_{i,t} + \delta X_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $\Delta y_{i,t}$ denotes the outcome of interest for country i in year t , for example, the change in sovereign spreads, and $\Delta P_{i,t}$ captures the change in the country's import or export prices.

Because countries in our sample are small relative to world markets, reverse causality is unlikely to bias the estimates: it is reasonable to view each country as a price taker. However, common global shocks could affect both commodity prices and sovereign spreads. For instance, a downturn in the global financial cycle may simultaneously depress commodity prices and raise spreads, inducing a negative bias in the OLS estimate of γ .

To address this concern, we implement a shift-share identification strategy. The design exploits cross-country heterogeneity in trade composition as a source of quasi-random

variation in exposure to global price innovations. We define the shift-share instrument as:

$$z_{i,t} = \sum_{j=1}^J \omega_{i,j,t-1} \varepsilon_{j,t}^P, \quad (2)$$

where $\omega_{i,j,t-1}$ is the share of commodity j in the total traded value (imports plus exports) of country i at time $t - 1$.

The denominator includes both imports and exports, so the shares do not sum to one. As a result, the instrument is a weighted sum rather than a weighted average of global price shocks. Following [Borusyak et al. \(2022\)](#), we control for the total trade weight $\Omega_{i,t} = \sum_{j=1}^J \omega_{i,j,t-1}$ to avoid bias from correlation between $\Omega_{i,t}$ and the instrument. This construction allows us to exploit variation in the net trade position of each country. For example, if a country is highly specialized in a few export commodities and imports little, a positive price shock to its export basket will have a stronger effect on its terms of trade than on its import prices, an effect that would be muted if import and export denominators were treated separately.

The shifts, $\varepsilon_{j,t}^P$, represent the innovations in the log change of the price of commodity j between $t - 1$ and t . We obtain these by modeling each commodity's log price change as an $\text{AR}(k_j)$ process and taking the residuals:

$$\Phi_{k_j}(L)\Delta p_{j,t} = \varepsilon_{j,t}^P. \quad (3)$$

This procedure ensures that the shocks represent the unanticipated component of global price changes. Intuitively, the 'surprise' component of price variation between $t - 1$ and t could not have influenced the lagged trade shares $\omega_{i,j,t-1}$, satisfying the exclusion restriction. The key identifying assumption, following [Borusyak et al. \(2022\)](#), is that conditional on fixed effects and controls, the shares $\omega_{i,j,t-1}$ are uncorrelated with unobserved country-level determinants of sovereign spreads. A potential concern is that trade specialization is not randomly assigned: countries that export oil or metals may differ systematically from commodity importers along dimensions such as institutional quality, political stability, or fiscal capacity, all of which independently affect default risk. We address this concern in three ways. First, the inclusion of country fixed effects absorbs all time-invariant differences across countries, so identification comes from within-country variation in commodity exposure over time rather than from cross-sectional differences in trade structure. Second, we control for real GDP growth, CPI inflation, and the exchange rate, which capture the main time-varying macroeconomic channels through which commodity prices could affect spreads independently of the shift-share instrument. Third, the leave-one-out exercise in [Figure 2](#) shows that no single country drives the results, which would be unlikely if the estimates reflected the idiosyncratic characteristics of a particular commodity-exporting or

commodity-importing economy.

Overall, the instrument captures exogenous cross-country variation in exposure to global price shocks. Countries differ in their trade composition, and these differences translate into heterogeneous exposure to unanticipated movements in world prices.

2.2 Data Description

The analysis covers 37 emerging and developing economies over the period 1991-2023.¹ We construct country-specific terms-of-trade indices using a set of 41 internationally traded commodities spanning energy products (crude oil, natural gas, coal), metals (copper, aluminum, iron ore, zinc), agricultural goods (wheat, rice, coffee, cocoa, soybeans), and livestock products (beef, pork, chicken, wool). The full list of commodities is reported in Table 8 in Appendix A. Commodity prices are drawn from the *IMF Primary Commodity Prices Database*, which provides consistent monthly series for a broad set of globally traded commodities. Prices for oranges, which are absent from the IMF database, are taken from the *World Bank Pink Sheet*. Trade weights are derived from bilateral trade flows reported in *UN Comtrade*.

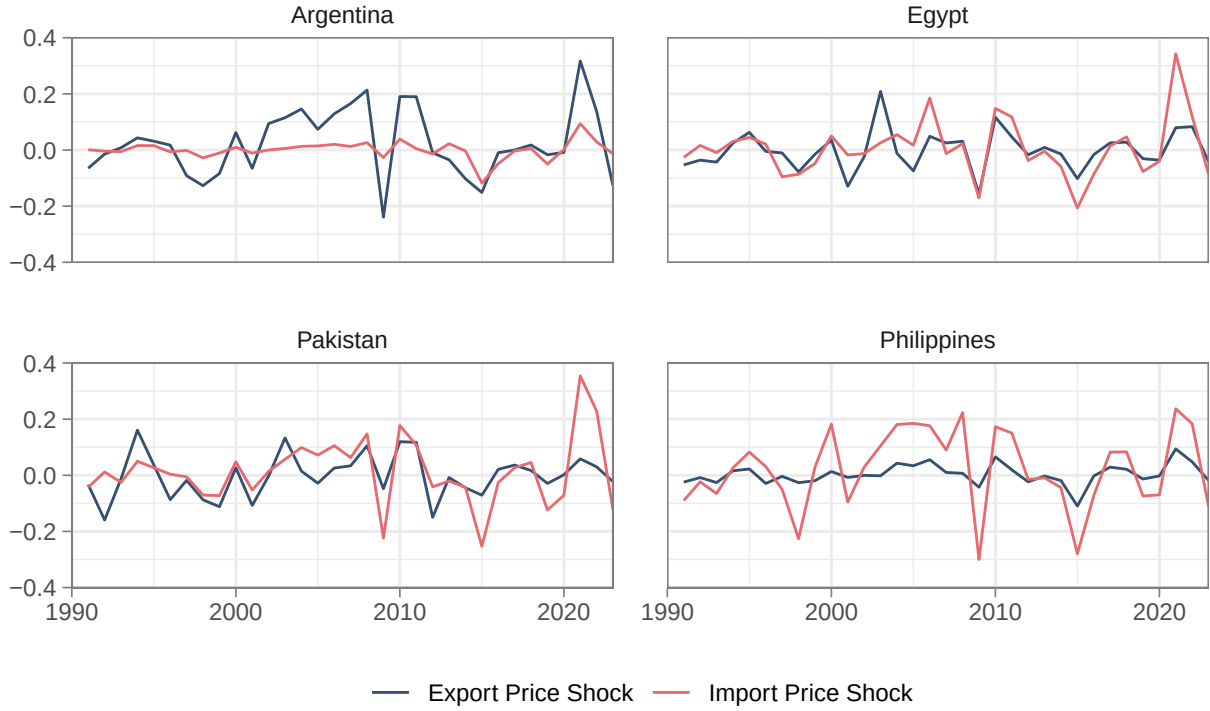
Real and nominal GDP are taken from the *IMF World Economic Outlook*, while external balance and debt indicators are obtained from the *IMF International Financial Statistics (IFS)* and the *IMF International Debt Statistics (IDS)*. Sovereign risk is measured using EMBI spreads from *Global Financial Database (GFD)* via Finaeon access.

2.3 Shift-Share Instrument

Figure 1 illustrates the shift-share instruments for import and export price shocks across countries. Identification stems from cross-country heterogeneity in trade exposure to global commodity price movements. In commodity-exporting economies, such as Argentina, fluctuations in world prices primarily translate into export-price shocks, whereas in import-dependent economies, such as the Philippines, the same global movements generate import-price shocks.

¹The dataset covers: Argentina, Belize, Brazil, Bulgaria, Chile, China, Colombia, Dominican Republic, Ecuador, Egypt, El Salvador, Gabon, Ghana, Hungary, Indonesia, Iraq, Jamaica, Kazakhstan, Lebanon, Malaysia, Mexico, Pakistan, Panama, Peru, Philippines, Poland, Russia, Serbia, South Africa, South Korea, Sri Lanka, Tunisia, Turkey, Ukraine, Uruguay, Venezuela, and Vietnam.

Figure 1: Shift-Share Instrument.



Note: The figure illustrates the shift-share instruments for import and export price shocks across countries. Identification stems from cross-country heterogeneity in trade exposure to global commodity price movements. In commodity-exporting economies such as Argentina, fluctuations in world prices primarily translate into export-price shocks, whereas in import-dependent economies such as the Philippines, the same global movements generate import-price shocks.

Since commodity prices exhibit substantial comovement, part of the variation is absorbed by time fixed effects. For instance, the global collapse in commodity prices in 2009 induces large negative shocks for both exporters and importers. However, residual price variation across commodities, combined with differences in countries' trade composition, continues to generate meaningful cross-country and time-series variation for identification. An illustrative example is the surge in cotton prices in 1994. Pakistan, where cotton exports accounted for nearly half of total commodity exports, experienced a pronounced positive export-price shock. In contrast, Argentina, Egypt, and the Philippines, where cotton exports were negligible, were largely unaffected.

2.4 Results

We now present the baseline estimates of the effects of import and export price shocks on sovereign risk, external borrowing, and the current account.

Sovereign spreads. Table 1 reports estimates for changes in EMBI spreads. A one-standard-deviation increase in import prices raises sovereign spreads by 77.1 basis points when entered alone (column 1), a large and statistically significant effect. Export price shocks, by contrast, are associated with a smaller and less precisely estimated effect of -41.2 basis points (column 2).

In the joint specification (column 3), the import price coefficient remains large and significant at 78.0 basis points, while the export price coefficient collapses to 1.18 basis points and becomes statistically insignificant, indicating that the negative univariate effect of export prices is not robust to controlling jointly for import price shocks. Including additional controls does not alter this result and further reinforces the effect of import price shocks.

These results are stable across robustness checks: restricting the sample to moderate spread changes below 500 basis points (column 5) preserves a significant import price effect of 30.5 basis points, while limiting to countries with external debt below 30% of GDP (column 6) yields a large and significant effect of 69.0 basis points.²

²The increase in R^2 from column (4) to column (5) reflects the mechanical compression of the dependent variable when extreme crisis episodes are excluded, rather than improved model fit; the qualitative results are unchanged.

Table 1: Baseline Results on EMBI Spreads.

	Δ Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	77.1*** (26.8)		78.0*** (23.8)	83.9*** (29.9)	30.5** (14.3)	69.0** (30.7)
Export Price Shock		-41.2* (24.0)	1.17 (20.1)	0.878 (26.5)	4.36 (9.61)	18.5* (9.33)
Observations	511	511	511	376	484	306
R ²	0.35	0.34	0.35	0.40	0.51	0.48
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports the estimated effects of import and export price shocks, measured using a shift-share instrument, on changes in EMBI spreads. All regressions include country and year fixed effects and control for the sum of the shares, real output growth, one-year CPI inflation, and the nominal exchange rate. Column (4) adds controls for average maturity, and the growth rates of interest and principal debt repayments. Column (5) restricts spread changes to less than 500 basis points. Column (6) restricts the external debt-to-output ratio to below 30%. Shocks are normalized to one standard deviation. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Heterogeneity by net trade position. The weak average effect of export price shocks masks substantial cross-country heterogeneity. A rise in export prices has two opposing consequences: it raises trade revenues, but also increases the domestic cost of goods that the country both exports and consumes. For a country that is a pure net exporter, the revenue effect dominates; for a country closer to balance, the two effects partially offset, muting the impact on spreads. To test this, we interact the export price shock with a time-invariant measure of each country's net trade position, constructed as the country-specific average of the difference between export and import share weights used in the shift-share instrument,

ensuring internal consistency with the shock construction.³

Table 2 reports the results. The interaction between the export price shock and the net trade position is negative and significant across all specifications, with a magnitude of 67.8 basis points in the joint baseline (column 3), robust to including more control variables (column 4), excluding extreme spread episodes (column 5) and high-debt countries (column 6). Countries with stronger net export positions thus experience significantly greater spread compression following export booms, while the effect is attenuated or absent for countries that both export and heavily consume the same commodities. The import shock interaction is positive in column (2) but loses significance in the joint specification (column 3), suggesting the heterogeneity is primarily driven by the export side.

³Since the net trade position index is time-invariant, it is absorbed by country fixed effects when entered alone; only its interaction with the time-varying shock is identified. This is standard in shift-share designs with time-invariant exposure measures.

Table 2: Net Trade Position Heterogeneity and EMBI Spreads.

	Δ Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	35.9 (23.4)	125.0*** (33.4)	71.4** (29.0)	71.6** (34.5)	39.0* (19.7)	54.2 (32.4)
Export Price Shock	77.6** (30.6)	2.03 (20.0)	71.5** (30.0)	71.1** (33.4)	31.0* (17.2)	52.5* (26.3)
Export Shock $\times$ Net Trade Position	-74.3*** (26.5)		-67.8** (25.1)	-79.0*** (24.9)	-25.1** (10.4)	-36.7* (20.9)
Import Shock $\times$ Net Trade Position		45.0** (21.8)	30.5 (19.0)	40.9 (24.9)	20.5 (14.5)	12.2 (22.6)
Observations	511	511	511	376	484	306
R ²	0.37	0.36	0.37	0.42	0.53	0.49
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table augments the baseline specification of Table 1 with interactions between the price shocks and a time-invariant index of net trade position, constructed as the country-specific average difference between export and import share weights from the shift-share instrument, normalized to unit standard deviation. Since the index is time-invariant, it is absorbed by country fixed effects; only the interaction with the time-varying shock is identified. All other controls and sample restrictions follow Table 1. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

External debt. Table 3 reports estimates for new external borrowing by the general government, normalized by nominal GDP. Both shocks are associated with negative coefficients throughout, consistent with price-driven tightening of financing conditions reducing new debt issuance. However, the estimates are imprecisely estimated in the univariate and joint baseline specifications (columns 1–3), and only reach marginal significance in the full-controls specification (column 4), where both import and export price shocks are associated with

declines of 0.34 and 0.25 percentage points of GDP respectively. The effects are close to zero and insignificant in the restricted subsamples (columns 5 and 6).⁴

Table 3: Baseline Results on New External Debt.

	Δ Debt					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	-0.260 (0.194)		-0.630 (0.391)	-0.337* (0.189)	0.014 (0.118)	0.045 (0.170)
Export Price Shock		-0.119 (0.168)	-0.455 (0.325)	-0.244* (0.133)	-0.138 (0.098)	0.038 (0.097)
Observations	755	755	755	755	351	520
R ²	0.37	0.37	0.38	0.61	0.57	0.42
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports estimated effects on new external debt issuance by the general government, normalized by nominal GDP. All other controls and sample restrictions follow Table 1. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Current account. Table 4 examines whether the spread and debt effects operate through the current account, providing a direct check on the real transmission of the shocks. Import price shocks reduce the current account balance by 1.7 percentage points of GDP when entered alone (column 1) and by 0.6 percentage points in the joint specification (column 3), both significant at conventional levels. Export price shocks have the symmetric effect, improving the current account by 1.7 percentage points alone (column 2) and 1.3 percentage points jointly (column 3). These results are robust to additional controls (column 4) and the low-debt subsample (column 6), though the export shock coefficient weakens in the low-

⁴The imprecision of the debt results is consistent with the spread evidence: higher borrowing costs tighten financing conditions, reducing debt issuance at the margin, but this quantity response is noisier and more heterogeneous across countries than the price response captured by spreads.

spread subsample (column 5), mirroring the pattern in Table 1. These findings serve two purposes. First, they validate the identification strategy: the shift-share instrument captures genuine terms-of-trade variation with measurable real effects before any sovereign risk channel is invoked. Second, they document a clear transmission chain: import price surges simultaneously erode the external balance and raise spreads, while export booms improve it and compress spreads, with the asymmetry in spread effects reflecting the partial offsetting of revenues and costs visible in Table 2.

Table 4: Baseline Results on the Current Account Dynamics.

	Δ Current Account (% GDP)					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	-1.71*** (0.463)		-0.621* (0.320)	-0.609* (0.326)	-1.97*** (0.511)	-1.23** (0.444)
Export Price Shock		1.67*** (0.447)	1.34** (0.522)	1.32** (0.518)	0.514 (0.372)	0.599* (0.294)
Observations	728	728	728	728	349	505
R ²	0.26	0.30	0.31	0.32	0.35	0.29
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports estimated effects on the annual change in the current account balance as a share of GDP. All other controls and sample restrictions follow Table 1. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Taken together, the four sets of results point to a coherent picture. Import price shocks simultaneously erode the current account, raise borrowing costs, and reduce debt issuance: three mutually reinforcing effects that generate large and precisely estimated increases in sovereign spreads. By contrast, export price shocks improve the external balance, but their effect on spreads is weak on average, as the associated revenue gains are partially offset by higher domestic costs in economies that consume what they export. This offset is consistent with the heterogeneity patterns documented in Table 2, where the spread-compressing effect

of export price increases is concentrated among net exporters.

These empirical findings suggest that import and export price shocks operate through two distinct but partially offsetting mechanisms. A rise in global commodity prices affects both the cost of living, through its impact on the domestic consumption basket, and real income, through its effect on trade revenues. For import-dependent economies, these two effects reinforce each other, compressing real income and tightening repayment capacity. For exporters, they partially offset, weakening the aggregate effect on sovereign risk.

The model developed in the next section formalizes this interpretation. It introduces global price shocks that simultaneously affect domestic prices and trade income, and allows us to quantify the relative importance of these two channels across different trade structures.

2.5 Robustness Checks

We assess the robustness of our results by excluding energy commodities from the construction of the shift-share instrument. This exercise serves two purposes. First, it addresses the concern that large exporters in the sample may influence global energy prices, potentially affecting identification. Second, it allows us to evaluate whether the baseline results are driven by energy commodities.

Table 5 reports the results for sovereign spreads. The import-price effect is attenuated in the univariate specification (column 1) and becomes statistically insignificant. However, it remains positive, large, and statistically significant in the joint specifications (columns 3 and 4), with magnitudes close to the baseline estimates (e.g., 66.7 basis points in column 4 compared to 83.9 in Table 1).

By contrast, excluding energy commodities materially changes the estimated effect of export price shocks. The coefficient becomes positive in all specifications and statistically significant throughout the joint specifications, including when the full set of controls is included (column 4). This contrasts with the baseline results, where the export effect is small and not robust, and indicates that the role of export price shocks depends importantly on the composition of traded commodities.

Overall, these results confirm that the import-price channel is robust across commodity groups, while the export-price channel is more sensitive to commodity composition. At the same time, once energy commodities are excluded, export price shocks display a positive and statistically significant effect on sovereign spreads, suggesting that the baseline attenuation partly reflects the distinct dynamics of energy prices. Additional results for external debt and current account outcomes, reported in Appendix A, show that quantity responses remain weaker and less precisely estimated than spread responses.

Table 5: Effect on Spreads, Excluding Energy Commodities.

	Δ Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	26.5 (20.9)		64.8*** (20.2)	66.7*** (22.0)	24.0** (9.36)	60.9** (24.9)
Export Price Shock		26.2* (13.9)	44.8*** (15.5)	38.0* (21.5)	19.0** (9.28)	42.4** (19.1)
Observations	511	509	509	376	482	306
R ²	0.34	0.34	0.36	0.40	0.52	0.49
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

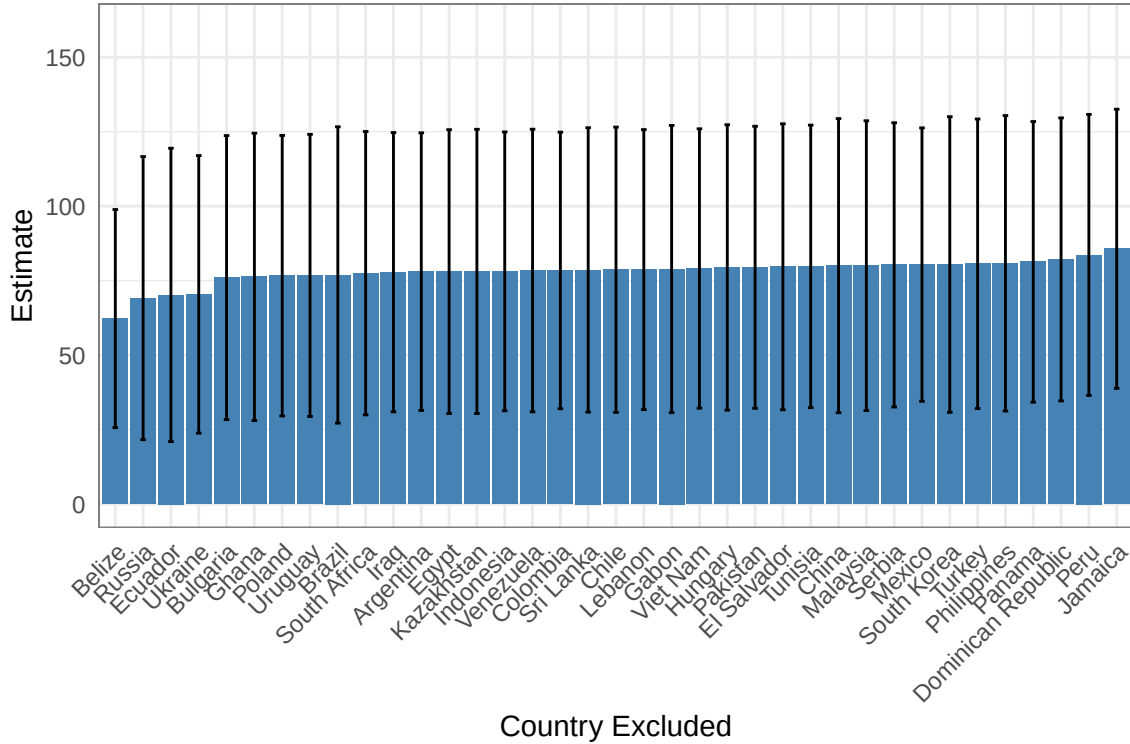
Note: This table reports the estimated effects of import and export price shocks, measured using a shift-share instrument constructed excluding energy commodities, on changes in EMBI spreads. The corresponding import and export share variables are computed using the same restricted set of commodities. All other controls and sample restrictions follow Table 1. Shocks are normalized to one standard deviation. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We next assess whether the results are driven by any single country in the sample. Figure 2 reports leave-one-out estimates of the effect of import price shocks on sovereign spreads, along with 95% confidence intervals. The specification corresponds to column (3) of Table 1, which includes both import and export price shocks as well as the baseline controls. The results are broadly stable across exclusions. While a small number of countries exert a noticeable influence on the magnitude of the estimates, most prominently Belize, Russia, Ecuador, and Ukraine, the overall pattern remains unchanged. In particular, excluding Belize reduces the estimated effect of import price shocks to around 50 basis points, but the coefficient remains economically large and statistically significant. Excluding other countries leads to only minor variations around the baseline estimate. Estimates for export price shocks display a similar degree of stability, as shown in Figure 11 in Appendix A. Results based on the full-controls specification (column (4) of Table 1) are also qualitatively unchanged, as

reported in Figures 12 and 13 in Appendix A.

Overall, these results indicate that the baseline estimates are not driven by any single influential country.

Figure 2: Leave-One-Out Estimates: Import Price Shocks.



Note: This figure shows the effect of import price shocks and sovereign spreads, excluding one country at a time from the sample. Each vertical bar corresponds to the country excluded. The horizontal brackets show the 95% confidence interval. The specification includes both import and export price shocks, and controls for the sum of the shares, the growth rate of real output, the one-year CPI inflation and the nominal exchange rate. Standard errors are clustered at the country level.

3 Model

Time is discrete and the horizon is infinite. We study a small open economy in which the government chooses borrowing and consumption while facing global price movements that simultaneously affect the cost of the consumption basket and the value of net exports, as suggested by the empirical evidence. International lenders are risk-neutral and price sovereign bonds competitively based on default risk; default triggers temporary market exclusion and a deterioration in the country's terms of trade.

Relative to [Arellano \(2008\)](#), the key innovation lies in the nature and transmission of shocks. Instead of exogenous income fluctuations, the economy is subject to *global price shocks* that affect both domestic absorption and the relative price of traded goods. These shocks operate through two structural mechanisms: a *price-index channel*, which alters the cost of the consumption basket, and a *terms-of-trade channel*, which changes national income through trade exposure. Depending on its trade structure, a positive global shock can be expansionary, easing borrowing conditions for an exporter, or contractionary, tightening credit for an importer.

3.1 Environment

Goods, Prices, and Global Price Shocks. Consider a small open economy with a continuum of tradable goods indexed by $i \in [0, 1]$.⁵ Each good i has a price $P_t(i)$ and a subsistence level $\bar{c}(i)$. Movements in world commodity markets are summarized by a set of K *global factors* capturing broad co-movements in international prices:

$$\ln P_t(i) = \mu(i) + a(i)^\top f_t + \varepsilon_t(i), \quad (4)$$

where $\mu(i)$ is the time-invariant mean of $\ln P_t(i)$; f_t collects the first K principal components of global commodity prices; $a(i)$ are the associated factor loadings capturing how each good co-moves with the *global factors*; and $\varepsilon_t(i)$ is an idiosyncratic residual. The factors follow a stationary VAR(1):

$$f_t = \Phi f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u). \quad (5)$$

Preferences and Aggregation. Households consume all goods according to a Stone-Geary aggregator:

$$c_t = \prod_{i \in [0, 1]} [c_t(i) - \bar{c}(i)]^{\lambda(i)}, \quad \int_0^1 \lambda(i) di = 1,$$

where $\bar{c}(i)$ denotes subsistence levels and $\lambda(i)$ the relative weight of each good in the consumption basket.

Budget Constraint and Trade. Let $e(i)$ denote endowments and D_t the period- t net resource inflow from the capital account (i.e. the flow version of the budget, given explicitly in equation (9) below). Aggregate expenditure satisfies:

$$E_t = \int_0^1 P_t(i) e(i) di + D_t = \int_0^1 P_t(i) c_t(i) di.$$

⁵In the quantitative implementation (Section 4), the continuum is discretized into the 41 IMF primary commodities included in the global-factor panel.

Define the net endowment profile $\tilde{e}(i) \equiv e(i) - \bar{c}(i)$, yielding the resource constraint:

$$E_t - \int_0^1 P_t(i) \bar{c}(i) di = D_t + \int_0^1 P_t(i) \tilde{e}(i) di. \quad (6)$$

Aggregate Consumption. Aggregating across goods yields:

$$c_t = \Gamma^c e^{-\mu^C - \psi^\top f_t} (D_t + \bar{Z} + \beta^\top f_t), \quad (7)$$

where $\Gamma^c \equiv \prod_i \lambda(i)^{\lambda(i)}$ normalizes the consumption index; $\mu^C \equiv \int_0^1 \lambda(i) \mu(i) di$ is the average log price level; $\psi \equiv \int_0^1 \lambda(i) a(i) di$ summarizes the basket's exposure to global prices (the *price-index exposure*); $\bar{Z} \equiv \int_0^1 \bar{P}(i) \tilde{e}(i) di$ measures baseline trade income evaluated at steady-state prices $\bar{P}(i) \equiv e^{\mu(i)}$; and $\beta_k \equiv \int_0^1 \bar{P}(i) \tilde{e}(i) a_k(i) di$ captures how the country's trade composition aligns with global factors (the *terms-of-trade exposure*). The trade-income term $\bar{Z} + \beta^\top f_t$ is a first-order Taylor expansion of $\int P_t(i) \tilde{e}(i) di$ around $f = 0$, obtained by linearizing $e^{a(i)^\top f_t}$; this linearization is the one implemented in the solver. A detailed derivation of this expression, including the price index, demand system, and intermediary steps, is provided in Appendix B.

Transmission channels. Global shocks affect consumption through two channels: (i) a *price-index effect* ($-\psi^\top f_t$), which raises the cost of consumption; and (ii) a *terms-of-trade effect* ($+\beta^\top f_t$), which shifts income via net exports. Holding external resources fixed, the elasticity of consumption with respect to factor $f_{k,t}$ is:

$$\frac{\partial \ln \widehat{c_t}}{\partial f_{k,t}} = -\psi_k + \underbrace{\frac{1}{\bar{D} + \bar{Z}}}_{\kappa_c} \beta_k.$$

Consumption increases when the income effect dominates the price effect, that is, when $\beta_k > (\bar{D} + \bar{Z})\psi_k$. The factor $\kappa_c = 1/(\bar{D} + \bar{Z})$ acts as an endogenous leverage term: economies with smaller external buffers, and thus tighter financial constraints, respond more strongly to global shocks. The expression above corresponds to the *steady-state* value of the leverage coefficient, evaluated at average debt $\bar{D}$. Its fully state-dependent counterpart, $\kappa_c(f, b) = 1/(D(f, b) + \bar{Z})$, used in Proposition 1 and the proofs, extends this expression by allowing leverage to vary with the endogenous resource flow.

Intuitively, when external resources are constrained, global price movements generate amplified fluctuations in real consumption and default risk. These forces discipline the government's default incentives in equilibrium. For net exporters of high-loading goods, favorable shocks raise consumption, reduce default risk, and compress sovereign spreads. By contrast, for import-dependent economies, the same shocks are stagflationary: they increase

the consumption price index, reduce real income, and raise default risk. These mechanisms are further developed in the next section.

3.2 Government Problem and Market Structure

A benevolent government maximizes expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta_d^t u(c_t), \quad (8)$$

where expectations are taken over realizations of the global shocks, $0 < \beta_d < 1$ is the discount factor, and $u(c)$ is a strictly increasing and concave period utility function. Aggregate consumption c_t is given by equation (7). The government's problem is subject to the aggregate resource constraint and the bond-market equilibrium conditions.

Timing, States, and Resource Flows. At the beginning of period t , the state of the economy is (f_t, b_t) , where f_t denotes the realization of the global factor and b_t the stock of sovereign debt carried from the previous period. After observing f_t , the government decides whether to *repay* or *default*. If it repays, it retains market access and chooses next-period debt b_{t+1} . Positive values of b_t correspond to net foreign assets, while negative values indicate outstanding debt.

Resource flows between the country and the rest of the world are:

$$D_t = b_t - q(f_t, b_{t+1}) b_{t+1}, \quad (9)$$

with $D_t > 0$ indicating net borrowing.

Risk-neutral international lenders price bonds competitively to satisfy zero expected profits:

$$q(f_t, b_{t+1}) = \frac{1 - \delta(f_t, b_{t+1})}{1 + r^*}, \quad (10)$$

where $\delta(f_t, b_{t+1})$ is the endogenous probability of default.

The implied sovereign spread is $s_t = 1/q(f_t, b_{t+1}) - (1 + r^*)$, expressed in gross one-period units. In the simulations, we annualize s_t as $(1 + s_t)^4 - 1$ to match the quarterly frequency of the global factor process.

In default, the economy enters financial autarky and re-enters credit markets with probability $\theta \in (0, 1)$.

3.3 Recursive Competitive Equilibrium

Given the state vector $s = (f, b)$, the government's policy $b'(f, b)$, the bond price $q(f, b')$, and the implied consumption $c(f, b)$, we define the equilibrium.

Value Functions. Let $V^r(f, b)$ and $V^d(f)$ denote the value functions under repayment and default, respectively:

$$V(f, b) = \max\{V^r(f, b), V^d(f)\}. \quad (11)$$

Under repayment, the government retains market access and optimally chooses b' . Let $\pi(f' | f)$ denote the transition density of f_{t+1} given $f_t = f$, obtained from the VAR(1) in equation (5). The repayment value satisfies:

$$V^r(f, b) = \max_{b'} \left\{ u(c) + \beta_d \int V(f', b') \pi(f' | f) df' \right\}, \quad (12)$$

$$c = \Gamma^c \exp(-\mu^C - \psi^\top f) \left[b - q(f, b') b' + \bar{Z} + \beta^\top f \right], \quad (13)$$

subject to a borrowing limit $b' \geq -\underline{b}$, which rules out Ponzi schemes and does not bind in equilibrium.

If the government defaults, it loses access to credit and faces costlier imports through higher effective prices for consumption goods. While in autarky, it cannot borrow or lend but regains access with probability $\theta \in (0, 1)$. The corresponding value is:

$$V^d(f) = u(c^d) + \beta_d \int [\theta V(f', 0) + (1 - \theta) V^d(f')] \pi(f' | f) df', \quad (14)$$

$$c^d = \chi \Gamma^c \exp(-\mu^C - \psi^\top f) (\bar{Z} + \beta^\top f), \quad (15)$$

where $\chi \in (0, 1)$ captures the consumption loss during default.

Default and Repayment Sets. Define the default and repayment regions as:

$$A(b) = \{f \in \mathbb{F} : V^r(f, b) \geq V^d(f)\}, \quad \mathcal{D}(b) = \{f \in \mathbb{F} : V^r(f, b) < V^d(f)\},$$

where $\mathbb{F}$ is the support of f . The probability of default implied by these regions is:

$$\delta(f, b') = \int_{\mathcal{D}(b')} \pi(f' | f) df'. \quad (16)$$

When $b' \geq 0$, default is impossible ($\delta = 0$); when $\mathcal{D}(b') = \mathbb{F}$, default is certain.

Definition 1. A *recursive competitive equilibrium* consists of functions for (i) consumption $c(f, b)$ and $c^d(f)$, (ii) debt policy $b'(f, b)$ and repayment and default sets $A(b)$ and

$\mathcal{D}(b)$, and (iii) bond prices $q(f, b')$, such that:

1. Given policies, $c(f, b)$ and $c^d(f)$ satisfy the intertemporal resource constraint.
2. Given $q(f, b')$, the policy $b'(f, b)$ and sets $A(b)$ and $\mathcal{D}(b)$ solve the government problem in (11)–(15).
3. Bond prices satisfy the zero-profit condition (10).

Discussion. The equilibrium jointly determines debt, default, and consumption as functions of the global state f . Global shocks affect sovereign spreads and default risk through the same mechanisms described earlier: a *price-index channel* (via ψ) and a *terms-of-trade income channel* (via β). Export-oriented economies benefit from favorable global shocks, while import-dependent ones face higher prices and tighter credit conditions. We now formalize how global shocks map into equilibrium consumption and default risk.

Proposition 1 (Global factors, consumption, and default incentives). *Let $f \in \mathbb{R}^K$ denote the vector of global price factors driving the economy, and let aggregate consumption in equilibrium be:*

$$c(f, b) = \Gamma^c \exp(-\mu^C - \psi^\top f) (D(f, b) + \bar{Z} + \beta^\top f),$$

where $D(f, b)$ is the equilibrium net resource flow consistent with policy and bond pricing. Define the consumption wedge:

$$\Delta_k(f, b) \equiv \frac{\partial \ln c(f, b)}{\partial f_k} = -\psi_k + \kappa_c(f, b) \beta_k, \quad \kappa_c(f, b) \equiv \frac{1}{D(f, b) + \bar{Z}}.$$

Then:

- (i) **Sign.** A marginal increase in factor f_k raises equilibrium consumption if $\Delta_k(f, b) > 0$ and lowers it if $\Delta_k(f, b) < 0$. The term $-\psi_k$ captures the price-index channel, while $\kappa_c(f, b)\beta_k$ captures the terms-of-trade income channel.
- (ii) **Amplification.** The leverage coefficient $\kappa_c(f, b)$ amplifies the income channel; smaller buffers imply stronger sensitivity.
- (iii) **Default incentives.** If $\Delta_k(f, b) > 0$, favorable shocks improve repayment incentives and compress spreads; if $\Delta_k(f, b) < 0$, they weaken repayment incentives and raise default risk.

Sketch of proof. Differentiate $\ln c(f, b)$ with respect to f_k , holding the debt policy b' fixed at its current optimum. Linearizing around $\bar{f} = \mathbf{0}$, the impact derivative decomposes into a price-index effect ($-\psi_k$) and a trade-income effect ($\kappa_c(f, b)\beta_k$). Applying the envelope theorem to $V^r(f, b) = \max_{b'} \{u(c(f, b, b')) + \beta_d \mathbb{E}[V(f', b') \mid f]\}$ shows that $\partial V^r / \partial f_k$ inherits

the sign of $\Delta_k(f, b)$, under positive persistence of f . The effect on default incentives then follows from a monotone single-crossing assumption on $V^r - V^d$ in f_k , which is verified numerically in all our calibrations. A detailed proof is provided in Appendix C.1. $\square$

Remark 1 (Exporter vs. importer exposure). *When $\beta_k > 0$, the country is a net exporter in goods with high loadings on factor k : global upturns raise income and reduce default risk. When $\beta_k < 0$, it is a net importer: similar shocks worsen the terms of trade and raise default risk. The relative magnitudes of ψ_k and $\kappa_c(f, b)\beta_k$ determine whether the shock is expansionary or contractionary.*

Extensions linking the consumption wedge $\Delta_k(f, b)$ to sovereign spreads (Corollary 1) and to global comparative statics under bounded leverage (Lemma 1), which establish how the qualitative results extend to large and persistent global shocks, are provided in Appendix C.2 and Appendix C.3, respectively. Lemma 1 shows that when either the *price-index* or the *terms-of-trade* channel dominates uniformly, the qualitative effects of global shocks extend beyond local approximations.

This result provides a robust bridge between the theoretical mechanism and the quantitative analysis that follows. In the calibrated model, countries differ primarily in their exposure parameter β , and the sign of Δ_k determines how large, persistent movements in global prices translate into macroeconomic and financial outcomes.

4 Quantitative Analysis

4.1 Calibration and Estimation Strategy

The model is calibrated at a quarterly frequency to match key features of global prices and sovereign borrowing in the data. The calibration builds directly on the empirical strategy in Section 2, which identifies the principal components of international commodity prices and country-specific exposure to global shocks.

Preferences and International Finance. Households (or the representative government) are assumed to have CRRA preferences:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad (17)$$

with risk aversion parameter $\sigma = 2$. The subjective discount factor is $\beta_d = 0.97$. The international risk-free rate is fixed at $r^* = 1\%$, and foreign creditors are risk-neutral. The probability of regaining market access after a default is $\theta = 0.1$, in line with typical values

for emerging economies, and the default penalty parameter is $\chi = 0.9$, reducing effective consumption during exclusion. To prevent Ponzi schemes, we impose a lower bound on sovereign debt, $b' \geq -\underline{b}$, where $\underline{b} = 0.9$ (so that $b' \geq -0.9$), assumed non-binding in equilibrium.

Global Price Factors. Following the empirical procedure described in Section 2, we use a principal component analysis (PCA) on the panel of global log commodity prices $\ln P_t(i)$ to extract the common factors that capture worldwide fluctuations in tradable goods prices. Specifically, the log price of each good is represented as:

$$\ln P_t(i) = \mu(i) + a(i)^\top f_t + \varepsilon_t(i), \quad (18)$$

where $\mu(i)$ is the average log price of good i , $a(i)$ the vector of factor loadings, and $\varepsilon_t(i)$ an idiosyncratic component.

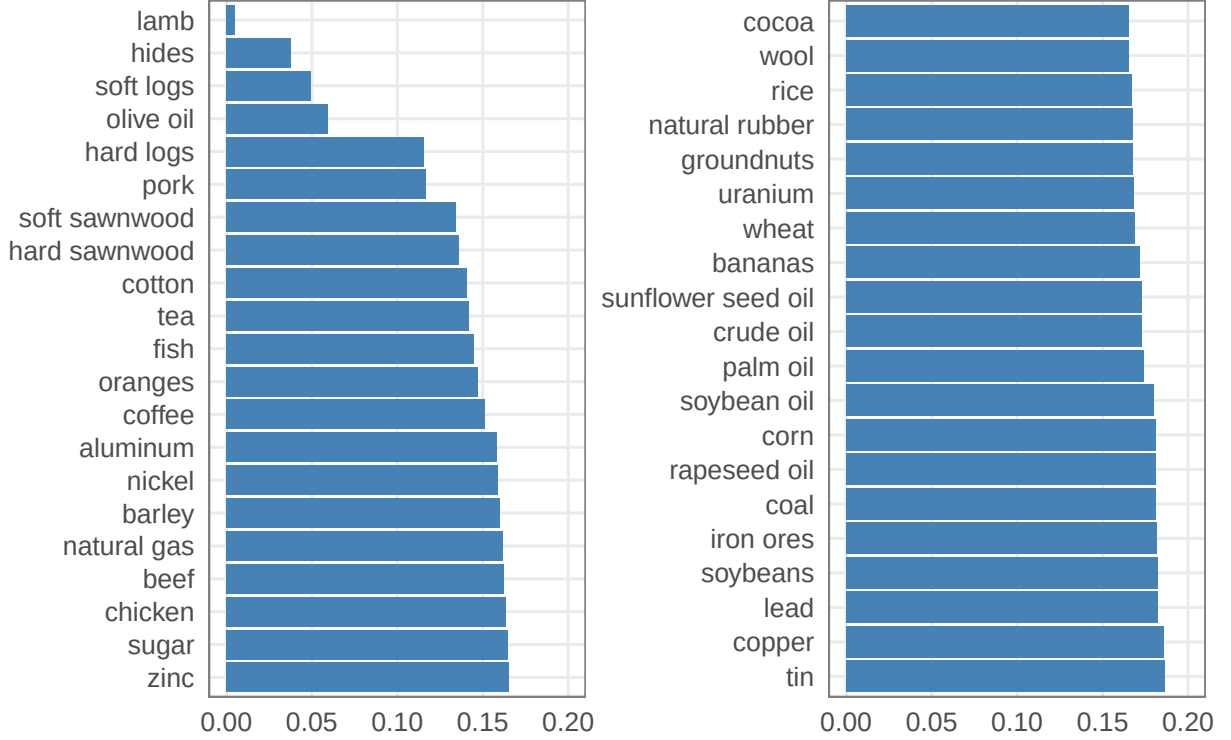
The empirical analysis shows that the first principal component explains more than 60% of the total variance in international prices, while the first five components together account for approximately 90% (see Figure 14 in Appendix A). Given the high explanatory power of the first component, we set the number of global factors in the model to $K = 1$. This choice captures the dominant common source of variation in international prices, interpreted as a global business-cycle or terms-of-trade factor. The factor follows a stationary autoregressive process:

$$f_t = \rho_f f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \sigma_f^2), \quad (19)$$

where $\rho_f = 0.97$ denotes persistence and $\sigma_f = 0.24$ the standard deviation of shocks. The process is estimated from the time series of the first principal component and discretized using the quadrature method of Tauchen & Hussey (1991). Details of the estimation procedure are provided in Appendix D.

Figure 3 reports the loadings of each commodity on the first principal component. Exposure to the global factor varies substantially across commodities. While energy products and minerals typically exhibit high loadings, a number of agricultural commodities are also strongly exposed, whereas others display much lower sensitivity. We exploit this heterogeneity in the quantitative analysis to construct trade structures with varying degrees of exposure to the global shock.

Figure 3: Factor Loadings.



Note: This figure shows the loadings of each commodity price on the first principal component.

Trade Structure and Exposure Parameters. The model requires two key exposure parameters: the consumption-price index exposure ψ and the net-export exposure β . These are computed as:

$$\psi = \int_0^1 \lambda(i) a(i) di, \quad \beta = \int_0^1 \bar{P}(i) \tilde{e}(i) a(i) di, \quad (20)$$

where $\lambda(i)$ denotes the consumption weights, $\tilde{e}(i) = e(i) - \bar{c}(i)$ the net-export profile of good i , and $\bar{P}(i) = \exp(\mu(i))$ its baseline price.

Cross-country heterogeneity in the model arises from differences in net endowments $\tilde{e}(i)$, which generate different values of the exposure parameter β across countries, and from differences in consumption baskets, which generate different values of ψ .

The steady-state trade term $\bar{Z}$ is given by:

$$\bar{Z} = \int_0^1 \bar{P}(i) \tilde{e}(i) di, \quad (21)$$

and the steady-state consumption price index constant is:

$$\mu^C = \int_0^1 \lambda(i) \mu(i) di. \quad (22)$$

Cross-country calibration of (ψ_i, β_i) . For the main quantitative exercise, we compute country-specific exposure parameters from observed trade and production data, rather than imposing stylized trade structures.

The key objects in the model are the net endowment profile $\tilde{e}(i) \equiv e(i) - \bar{c}(i)$, baseline prices $\bar{P}(i) \equiv e^{\mu(i)}$, and consumption weights $\lambda(i)$. Together, these determine the exposure parameters $\bar{Z}$, β , and ψ (equation (7)). In the data, $\bar{P}(i)\tilde{e}(i)$ corresponds to the value of net exports of commodity i , that is, production in excess of domestic consumption evaluated at international prices. Similarly, $\lambda(i)$ corresponds to the share of commodity i in the country's consumption basket.

For each country i and commodity $j \in \{1, \dots, 41\}$, let X_{ij} and M_{ij} denote the average values of exports and imports over the sample period, and let V_{ij}^P denote the average value of domestic production, all measured in USD. The empirical counterpart of $\bar{P}(j)\tilde{e}_i(j)$ is net exports, $NX_{ij} \equiv X_{ij} - M_{ij}$. The counterpart of $\lambda_i(j)$ is the share of commodity j in apparent domestic consumption, defined as $C_{ij} \equiv \max\{V_{ij}^P - X_{ij} + M_{ij}, 0\}$.

The discrete-sum analogues of the model's integral definitions are then given by:

$$\bar{Z}_i = \sum_{j=1}^{41} NX_{ij}, \quad \beta_i = \sum_{j=1}^{41} NX_{ij} a(j), \quad \lambda_{ij} = \frac{C_{ij}}{\sum_k C_{ik}}, \quad \psi_i = \sum_{j=1}^{41} \lambda_{ij} a(j), \quad (23)$$

where $a(j)$ is the loading of commodity j on the first principal component of the log-price panel, which is the same factor that enters the model's price equation (4).

Intuitively, β_i is the discrete counterpart of $\int \bar{P}(i)\tilde{e}(i) a(i) di$: it measures how strongly a country's trade income covaries with the global commodity factor. A country that net-exports commodities with high loadings $a(j)$ (such as tin, copper, lead, or soybeans) has a large positive β_i , implying that global price booms raise its trade income and, through the terms-of-trade channel, its real consumption. Conversely, a country that net-imports these goods has a negative β_i .

The parameter ψ_i is the counterpart of $\int \lambda(i) a(i) di$: it captures how the country's consumption price index responds to the global factor. A country that consumes predominantly high-loading commodities has a large ψ_i and therefore faces a stronger price-index drag when the global factor rises.

The interplay between these two channels, the income channel (β_i) and the price-index channel (ψ_i), determines whether global expansions are, on balance, beneficial or detrimental

for the sovereign, as formalized in Proposition 1.

To make β_i dimensionless and comparable to ψ_i , we normalize it by the country’s total net trade income $\bar{Z}_i$, defining $\tilde{\beta}_i \equiv \beta_i/|\bar{Z}_i|$ and setting $\bar{Z} \equiv 1$ in the model. The resulting parameters $(\psi_i, \tilde{\beta}_i)$ are then passed to the solver on a country-by-country basis.

The full procedure, including data sources, normalization choices, and a discussion of outliers, is described in Appendix E. Country-specific parameter values and the resulting simulated moments are reported in Appendix F. All subsequent quantitative results use these country estimates.

Parameter Summary. Table 6 summarizes the parameter values used in the quantitative exercises.

Table 6: Model Parameters and Exposure Estimates.

Parameter	Symbol	Value
Risk aversion	σ	2
Discount factor	β_d	0.97
Risk-free rate	r^*	1%
Persistence of global factor	ρ_f	0.97
Volatility of global factor	σ_f	0.24
Re-entry probability	θ	0.10
Default penalty	χ	0.90
Steady-state trade term	$\bar{Z}$	1
Debt bounds	b	$[-0.9, 0.9]$
Exposure parameters		
Argentina	$(\psi, \tilde{\beta})$	(0.168, 0.171)
Philippines	$(\psi, \tilde{\beta})$	(0.158, -0.167)

Note: The debt grid consists of 201 equally spaced points on $[-0.9, 0.9]$, and the global factor process f_t is discretized into 60 nodes using the quadrature method of Tauchen & Hussey (1991) with half-width $4\sigma_f$. Country-specific exposure parameters $(\psi_i, \tilde{\beta}_i)$ are constructed from trade and production data as described in Appendix E; full estimates and simulated moments are reported in Appendix F.

Numerical Implementation. Given $(\psi, \beta, \bar{Z}, \mu^C)$, we solve for the recursive equilibrium using value function iteration and the lenders’ zero-profit condition. Appendix D provides details on the estimation of exposure parameters and the full numerical procedure.

4.2 Quantitative Results

Using the calibration from Section 4.1, we solve the model separately by country, with (ψ_i, β_i) computed from country-specific trade and consumption shares. For each country, we simulate 500 histories of length 10,000 periods and examine the equilibrium relationship between global commodity prices, sovereign spreads, and default risk.

To illustrate the mechanism underlying this cross-country heterogeneity, we focus on two countries with sharply contrasting exposures. Argentina is the country most strongly affected by export-side commodity shocks in the empirical analysis (Figure 1): the model classifies it as an exporter-type economy, with sovereign spreads that are countercyclical with the global commodity factor. The Philippines, by contrast, is most strongly affected by import-side shocks and is classified as an importer-type economy, with procyclical spreads. We first present the equilibrium default sets, bond price schedules, and simulated moments for these two countries, before turning to the full cross-country distribution. The complete set of results for all 37 countries (including default heatmaps, bond price schedules, simulated moments, and default-event dynamics) is reported in Appendix F (Table 14 and Figures 15-23).

4.2.1 Default Incentives and Bond Prices

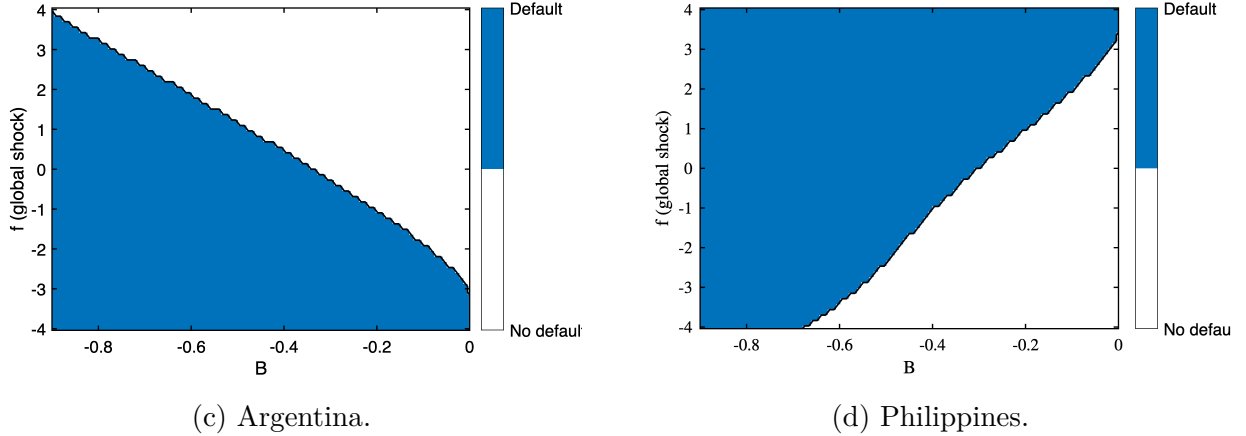
Figure 4 presents the default sets for Argentina and the Philippines. In both cases, higher levels of indebtedness increase default incentives, as more resources are freed for consumption upon default. However, the effect of the global factor differs sharply across the two economies.

In Argentina (Panel c), an increase in the global factor reduces default incentives. Argentina is a net exporter of commodities with high loadings on the global factor, so global price increases improve its terms of trade and raise trade income. At the same time, higher global prices increase the domestic cost of consumption. In the calibration, these two channels, the income (terms-of-trade) channel and the price-index channel, operate in opposite directions, but the income effect dominates. As a result, real consumption rises when the global factor increases. Since default is optimal when marginal utility is high, that is, when consumption is low, higher global factors reduce the incentive to default. Default incentives are therefore strongest when the global factor is low, as falling export prices depress income and consumption.

In the Philippines (Panel d), by contrast, default incentives increase with the global factor. The Philippines is a net importer of commodities with high global exposure, so a global price increase both worsens its terms of trade and raises the domestic consumption price index. In this case, the income and price-index channels reinforce each other, both reducing real consumption when the global factor rises. This amplification raises marginal utility and, in turn, default incentives.

Consistent with Proposition 1, the sensitivity of default incentives to the global factor is stronger in Argentina than in the Philippines. The default frontier is therefore steeper in the exporter case, reflecting the fact that the opposing channels generate larger state-dependent variation in real consumption.

Figure 4: Default Sets.



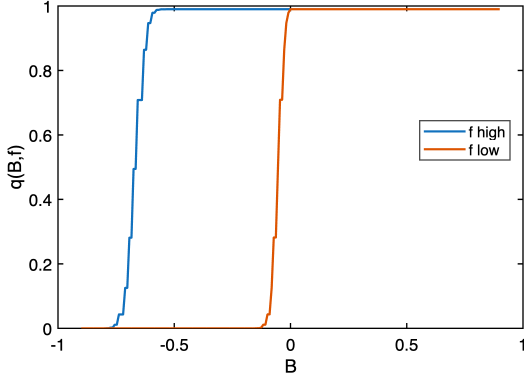
Note: Each panel shows the combinations of debt b and global factor f for which default is optimal (shaded region). Default sets for the remaining 35 sample countries are reported in Figures 15-18 of Appendix F.

In equilibrium, default incentives are reflected in the bond price schedule, shown in Figure 5. The figure highlights how the global factor affects borrowing conditions differently across countries through the interaction of the income and price-index channels.

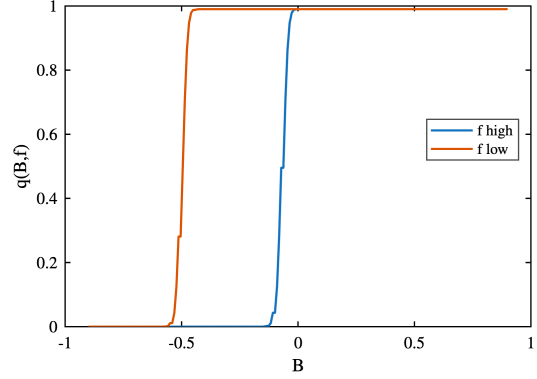
In Argentina, an increase in the global factor raises bond prices and reduces sovereign spreads. As a net exporter of high-loading commodities, Argentina benefits from an improvement in trade income when global prices rise. Although higher prices also increase the cost of consumption, the income effect dominates, lowering default risk and increasing bond prices.

By contrast, in the Philippines, higher global factors lower bond prices and increase spreads. As a net importer, the Philippines experiences both a deterioration in trade income and an increase in the domestic price index when global prices rise. The two channels therefore reinforce each other, reducing real consumption, increasing default risk, and compressing bond prices.

Figure 5: Bond Price Schedules.



(c) Argentina.



(d) Philippines.

Note: Each panel shows the equilibrium bond price $q(f, b')$ as a function of next-period debt b' , for different values of the global factor f . Bond price schedules for the remaining 35 sample countries are reported in Figures 19-22 of Appendix F.

4.2.2 Dynamic Transmission of Global Shocks

We now examine the dynamic properties of the model and the propagation of global shocks. For each country, we simulate 500 economies over 10,000 periods using the country-specific $(\psi_i, \tilde{\beta}_i)$ calibration. Table 7 reports the main simulated moments for the two illustrative countries; while the remaining 35 sample countries are reported in Table 14 in Appendix F.

Under the Argentine calibration, the model generates a mean debt level of about 0.23, relative to an average consumption normalized close to 1. Under the Philippine calibration, the mean debt level is lower, around 0.14, reflecting tighter borrowing conditions in an importer-type economy with more volatile consumption. These debt levels can be sustained in equilibrium because default entails a direct consumption cost, and because the high volatility of the global factor strengthens the sovereign's incentive to smooth consumption intertemporally, making autarky more costly.

Cross-country comparison. Four patterns emerge, all consistent with the interaction between the terms-of-trade (income) channel and the price-index channel.

(i) *Opposite cyclicalities of spreads.* The covariance between spreads and the global factor flips sign across the two economies, from -1.11 in Argentina to $+0.94$ in the Philippines. This sharp reversal is the central prediction of Proposition 1. For Argentina, a positive global shock raises trade income through the terms-of-trade channel, lowering default risk and compressing spreads. For the Philippines, the same shock raises the domestic price index and reduces real income, increasing default risk and widening spreads.

Table 7: Simulation Moments Across Main-text Calibrations.

Country	Annual Spread (%)			Debt			Consumption		
	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
Argentina	0.70	3.78	-1.11	0.231	0.113	0.109	0.980	0.035	0.014
Philippines	0.46	3.28	0.94	0.135	0.149	0.025	1.038	0.343	-0.345

Note: Each row reports simulation moments for one country over 500 runs of 10,000 periods. Annual Spread is in percent. Debt and Consumption are in model units with $\bar{Z}_i \equiv 1$. Cov with F is the sample covariance with the global factor. The global factor itself has Mean ≈ 0.002 and Std ≈ 1.018 across the two calibrations. The analogous table for all 37 sample countries is reported in Table 14 of Appendix F.

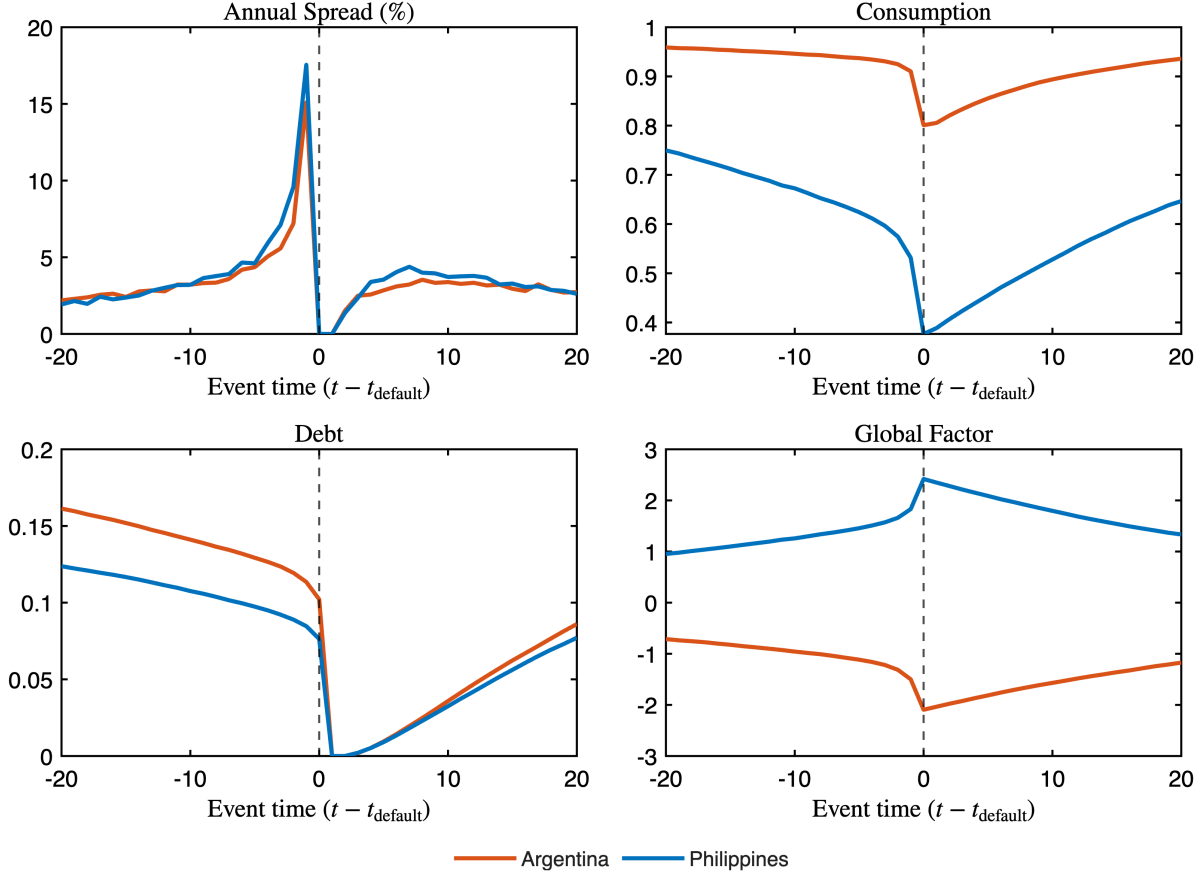
(ii) *Amplification of consumption volatility in importer economies.* Consumption volatility is an order of magnitude larger in the Philippines than in Argentina, with standard deviations of 0.343 and 0.035, respectively. The covariance with the global factor also flips sign, from essentially zero in Argentina (+0.014) to strongly negative in the Philippines (−0.345). This reflects the interaction of the two channels: in Argentina, the income and price-index effects offset each other, dampening fluctuations, whereas in the Philippines they reinforce each other, amplifying the decline in real consumption following positive global shocks.

(iii) *Higher borrowing capacity for exporter-type economies.* Argentina sustains a substantially higher mean debt level than the Philippines (0.23 of consumption versus to 0.14). Despite this, its mean spread is also higher (0.70% versus 0.46%), reflecting a stronger sensitivity of default risk to global conditions. The key mechanism is that favorable global shocks relax Argentina’s borrowing constraint precisely when the sovereign is most exposed, allowing it to sustain higher debt without proportionally increasing default risk.

(iv) *Dominance of tail risk in spread dynamics.* In both economies, spread volatility far exceeds its mean, with standard deviations of 3.78% and 3.28%. Sovereign risk is therefore driven primarily by tail realizations of the global factor rather than average conditions, consistent with the empirical concentration of spread spikes during crisis episodes.

We next examine the dynamics around default events. Figure 6 shows the average behavior of key variables around defaults across simulations. In both economies, defaults are triggered by extreme realizations of the global factor, but the direction differs across types. In the Philippines, defaults follow large positive shocks, as the price-index and terms-of-trade channels jointly compress real income. In Argentina, defaults follow negative shocks that reduce export revenues; the associated decline in consumption is more muted, as the price-index channel partly offsets the income loss. In both cases, default episodes are accompanied by sharp increases in spreads, with larger spikes in Argentina, reflecting the stronger sensitivity of its default incentives to global shocks.

Figure 6: Dynamics Around Default Events.



Note: Each line traces the average dynamics of a key variable around a default event for one country (Argentina in red, Philippines in blue). Default events are drawn from 500 simulations of 10,000 periods per country. Figure 23 in Appendix F reports the corresponding dynamics for the remaining 35 sample countries.

4.2.3 Cross-Country Heterogeneity and Model Validation

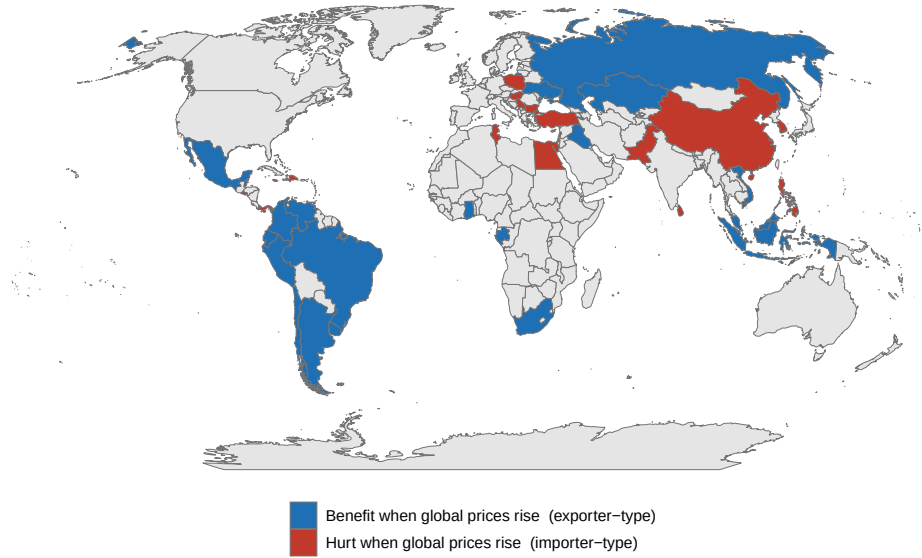
The Argentina–Philippines comparison highlights the core mechanism of the model in two polar cases. We show that the mechanism extends to the full sample and explains systematic cross-country heterogeneity in sovereign risk. Figure 7 summarizes the central prediction of the model for all 37 countries. For a given increase in global commodity prices, the response of sovereign risk depends on the relative strength of two forces: the terms-of-trade income channel and the price-index channel.

In 20 countries, the income channel dominates. Higher global prices raise trade income, increase real consumption, and reduce default risk, leading to a compression of sovereign spreads. These countries, shown in blue, include major commodity exporters such as South Africa, Chile, Mexico, Peru, Argentina, and Brazil, as well as economies where the income

channel only slightly outweighs the price-index effect, such as Colombia, Ecuador, Gabon, Indonesia, and Russia.

In the remaining 17 countries, the price-index channel dominates. Higher global prices increase the cost of the consumption basket, reduce real income, and raise default risk, leading to an increase in spreads. These countries, shown in red, are primarily manufacturing-oriented economies with large net commodity imports, including China, Pakistan, Poland, Bulgaria, Turkey, and Hungary.

Figure 7: Simulated Response of Sovereign Spreads to the Global Commodity Factor.



Note: Colours summarize the model's predictions based on country-specific exposure parameters $(\psi_i, \tilde{\beta}_i)$. Blue denotes countries where a global commodity expansion reduces default risk, while red denotes countries where it increases default risk. Grey indicates countries outside the sample. Full results are reported in Appendix F.

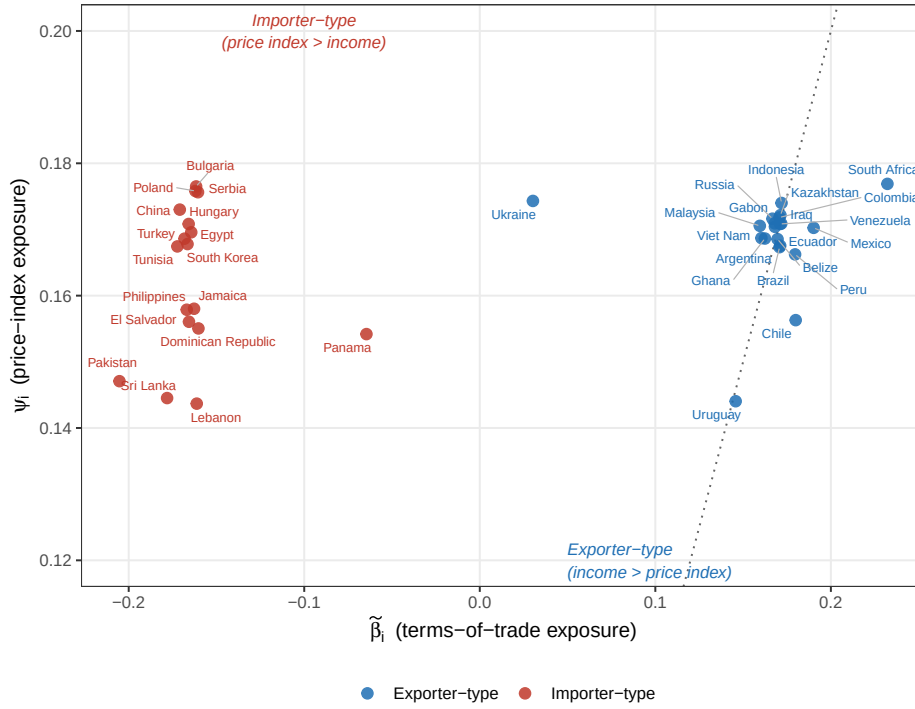
These patterns raise two questions: whether the estimated exposure parameters $(\tilde{\beta}_i, \psi_i)$ generate meaningful cross-country variation in the relative strength of the two channels, and whether the model's predictions for spread cyclicity are consistent with the data.

We proceed in three steps. First, we characterize the cross-country distribution of exposure parameters. Second, we compare model-implied and empirical spread cyclicity. Third, we assess whether the timing of default events is consistent with the model's mechanism.

Exposure parameters across countries. Figure 8 plots the calibrated exposure pairs $(\tilde{\beta}_i, \psi_i)$. The 45-degree line marks the threshold at which the two channels exactly offset each other. Countries below this line are dominated by the income channel, while those above it are dominated by the price-index channel.

The distribution reveals substantial heterogeneity. Commodity exporters such as South Africa, Chile, Mexico, and Peru lie well below the threshold, with large positive $\tilde{\beta}_i$ and moderate ψ_i . Manufacturing-oriented importers such as China, Turkey, Hungary, and Poland lie above the threshold, with negative $\tilde{\beta}_i$ and higher ψ_i . Several countries, including Argentina, Brazil, Colombia, and Ecuador, lie close to the threshold, reflecting a near offset between the two channels. Ukraine and Panama lie close to the vertical axis where $\tilde{Z}_i \approx 0$, implying unstable normalized exposure coefficients $\tilde{\beta}_i$. Their simulated covariances with the global factor are correspondingly close to zero, so they are best interpreted as near-threshold cases rather than as clearly classified economies.

Figure 8: Cross-Country Exposure Parameters.



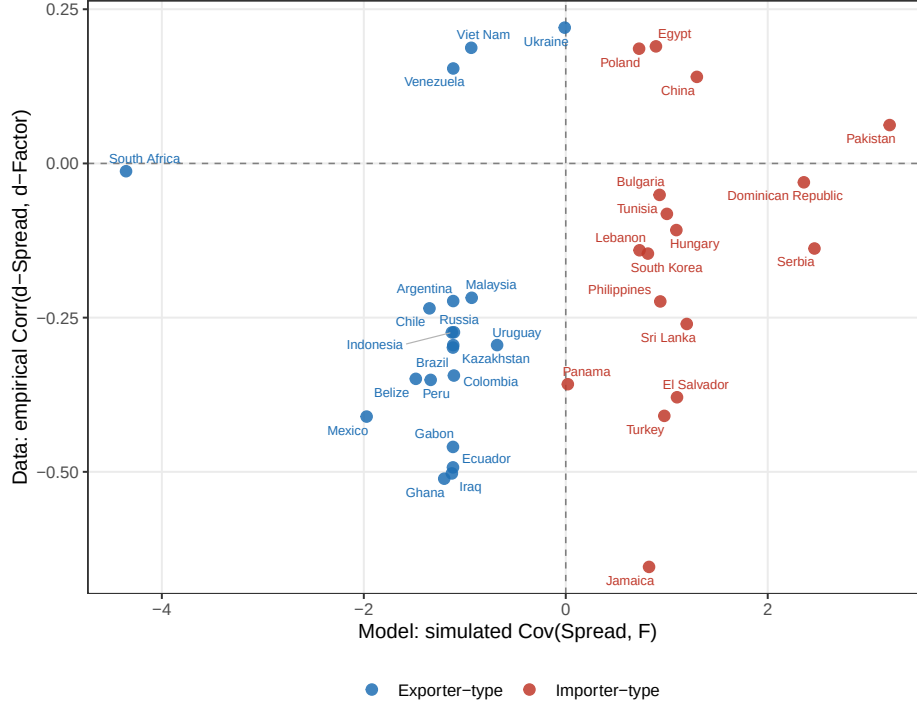
Note: Each point represents a country located at its calibrated $(\tilde{\beta}_i, \psi_i)$. The dotted line marks the threshold $\psi_i = \tilde{\beta}_i$. Countries below the line are exporter-type, while those above are importer-type.

Model vs. Data. Figure 9 compares the model-implied spread cyclicality, measured by $\text{Cov}(\text{Spread}, F)$, with the empirical correlation between EMBI spreads and the global commodity factor.

The model captures the sign of spread cyclicality for the majority of countries. Exporter-type economies cluster in the lower-left quadrant, where both the model and the data indicate that spreads fall when global prices rise. Importer-type economies display greater dispersion, reflecting additional forces in the data such as global risk appetite and domestic conditions

that are not modeled. Nevertheless, the sign pattern predicted by the model is broadly consistent with the data.

Figure 9: Model Prediction vs. Data: Spread Cyclicity.

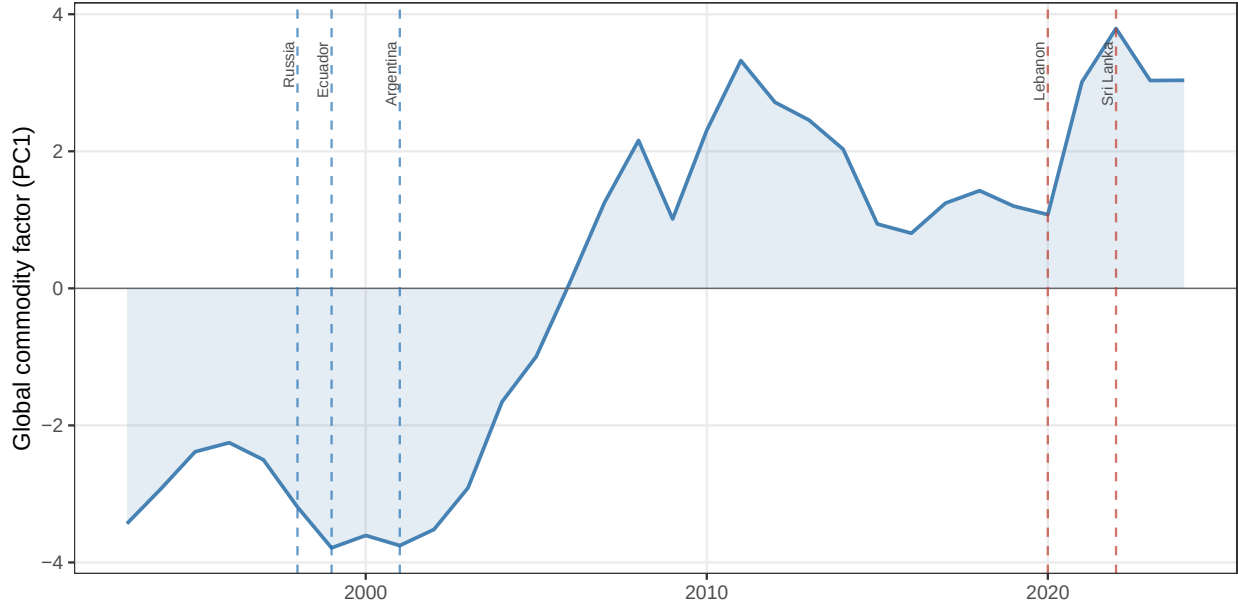


Note: The horizontal axis reports simulated $\text{Cov}(\text{Spread}, F)$ from the model. The vertical axis reports the empirical correlation between EMBI spreads and the global commodity factor over 1993–2024. Correct predictions correspond to observations in the lower-left and upper-right quadrants.

Historical validation. Figure 10 plots the global commodity factor together with selected sovereign default episodes. The factor captures well-known commodity cycles, including the late-1990s downturn, the 2003–2008 boom, the 2008 collapse, and the recent recovery.

The timing of defaults is consistent with the model’s mechanism. Exporter-type defaults, such as Russia (1998), Ecuador (1999), and Argentina (2001), occur during troughs in the global factor, when trade income collapses. Importer-type defaults, such as Lebanon (2020) and Sri Lanka (2022), occur during periods of high or rising global prices, when the cost of consumption increases. While other factors are clearly at play, this alignment supports the role of the two channels in shaping default risk.

Figure 10: Global Commodity Factor and Sovereign Default Episodes.



Note: The shaded area shows the first principal component of commodity prices (1993–2024). Vertical lines mark default events, with blue for exporter-type countries and red for importer-type countries.

Link to the empirical evidence. The quantitative results provide a structural interpretation of the reduced-form findings in Section 2. The shift-share regressions show that import price shocks raise spreads while export price shocks reduce them. In the model, this asymmetry arises from the interaction between the price-index and terms-of-trade channels. Countries with high ψ_i relative to $\tilde{\beta}_i$ experience a tightening of financial conditions following price increases, while those with high $\tilde{\beta}_i$ benefit from improved trade income. The cross-country classification therefore maps directly into the empirical pattern of spread responses.

Robustness. The sign of spread cyclicity is robust to changes in key parameters. Varying the default penalty χ , risk aversion σ , and re-entry probability θ does not alter the classification of countries into exporter-type or importer-type. While the magnitude of spreads and default frequencies changes, the sign of the response remains governed by the trade structure $(\psi_i, \tilde{\beta}_i)$, confirming that cross-country heterogeneity is driven primarily by exposure to global price shocks rather than by preference or institutional parameters.

5 Conclusion

This paper shows that trade structure is a first-order determinant of how global commodity price swings translate into sovereign risk. The same global shock that compresses spreads for a copper exporter widens them for an economy that imports the same metal, not because fundamentals differ, but because the balance between cost-of-living and income effects depends entirely on which side of the trade the country sits on.

Empirically, a shift-share design based on detailed import and export baskets reveals a sharp asymmetry: import price shocks raise sovereign spreads by around 70–80 basis points per standard deviation and erode the current account, while export price shocks have weak average effects that mask substantial heterogeneity driven by net trade position.

Theoretically, this paper formalizes this asymmetry within a sovereign default model in which a global commodity factor operates through two competing channels: a *price-index channel* that raises the cost of consumption and a *terms-of-trade income channel* that shifts trade revenues. Their relative strength defines a consumption wedge that governs whether a global upturn is expansionary or contractionary for the sovereign.

Quantitatively, solving the model for 37 emerging economies with country-specific calibrations delivers a clean empirical test: the sign of spread cyclicalities flips at the theoretical threshold with no country on the wrong side of zero, the predicted sign matches EMBI data for the vast majority of the sample, and historical default episodes, from Russia 1998 and Argentina 2001 to Lebanon 2020 and Sri Lanka 2022, align with the model’s predictions about when crises occur relative to global commodity cycles.

Three implications follow. First, sovereign risk assessment should incorporate trade structure as a primitive: the same global shock calls for opposite risk signals depending on whether the country exports or imports the goods driving global prices. Second, international support mechanisms (IMF facilities, contingent credit lines, commodity hedging programs) should be calibrated to the direction of exposure, not just its magnitude; a framework designed for exporters may be precisely wrong for importers facing the same global shock. Third, the leverage term in the model suggests that countries with thinner fiscal buffers face disproportionately amplified transmission, pointing to precautionary buffer accumulation as a priority for the most exposed economies. In a world of increasingly synchronized commodity cycles, recognizing that the same price movement can be simultaneously a tailwind and a headwind, depending on where a country sits in global trade networks, is essential for sovereign debt management and international financial stability.

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Appendix

A Empirical Appendix

Table 8 lists the 41 internationally traded commodities used to construct the country-specific terms-of-trade indices. Commodity prices are taken from the IMF Primary Commodity Prices Database, with the exception of oranges, for which we use the World Bank Pink Sheet series. Factor loadings on the first principal component are reported in Figure 3 of the main text.

Table 8: List of Commodities by Category.

Energy	Metals & Minerals	Agriculture	Other Primary
Crude oil	Aluminum	Bananas	Cotton
Natural gas	Copper	Barley	Hard logs
Coal	Iron ore	Beef	Soft logs
	Lead	Chicken	Hard sawnwood
	Nickel	Cocoa	Soft sawnwood
	Tin	Coffee	Hides
	Uranium	Corn	Natural rubber
	Zinc	Fish	Wool
		Groundnuts	
		Lamb	
		Olive oil	
		Oranges	
		Palm oil	
		Pork	
		Rapeseed oil	
		Rice	
		Soybean oil	
		Soybeans	
		Sugar	
		Sunflower oil	
		Tea	
		Wheat	

The following Tables 9, 10 and 11 report additional robustness results based on the shift-share instrument constructed excluding energy commodities. They cover heterogeneity by

net trade position, as well as effects on external debt and the current account.

Compared with the main specification in Table 3, the results are qualitatively similar. In most specifications, both import and export price shocks are associated with negative but statistically insignificant effects on new external debt. When the sample is restricted to low-debt or low-spread countries, the estimated effects are close to zero. Overall, these results reinforce the conclusion that quantity responses are weaker and less precisely estimated than the corresponding price responses in sovereign spreads.

Table 9: Net Trade Position Heterogeneity and EMBI Spreads,
Excluding Energy Commodities.

	Δ spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	45.1** (20.1)	58.0** (21.5)	47.4** (21.4)	50.4* (25.8)	19.2 (12.0)	68.4* (37.4)
Export Price Shock	75.1*** (22.6)	45.1*** (15.4)	76.2*** (23.1)	76.8*** (26.5)	31.0** (13.2)	16.8 (9.92)
Observations	509	509	509	376	482	306
R ²	0.36	0.36	0.36	0.41	0.52	0.48
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports the estimated effects of import and export price shocks, measured using a shift-share instrument constructed excluding energy commodities, on changes in EMBI spreads. The specification augments Table 5 with interactions between the shocks and the net trade position. All other controls and sample restrictions follow Table 5. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Effect on New External Debt, Excluding Energy Commodities.

	Δ debt					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	-0.169 (0.164)		-0.279 (0.247)	-0.179* (0.102)	0.031 (0.087)	0.095 (0.100)
Export Price Shock		-0.186 (0.213)	-0.261 (0.264)	-0.088 (0.085)	-0.078 (0.106)	-0.040 (0.088)
Observations	755	755	755	755	351	520
R ²	0.37	0.37	0.38	0.61	0.57	0.42
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports estimated effects on new external debt issuance by the general government, normalized by nominal GDP. All other controls and sample restrictions follow Table 5. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

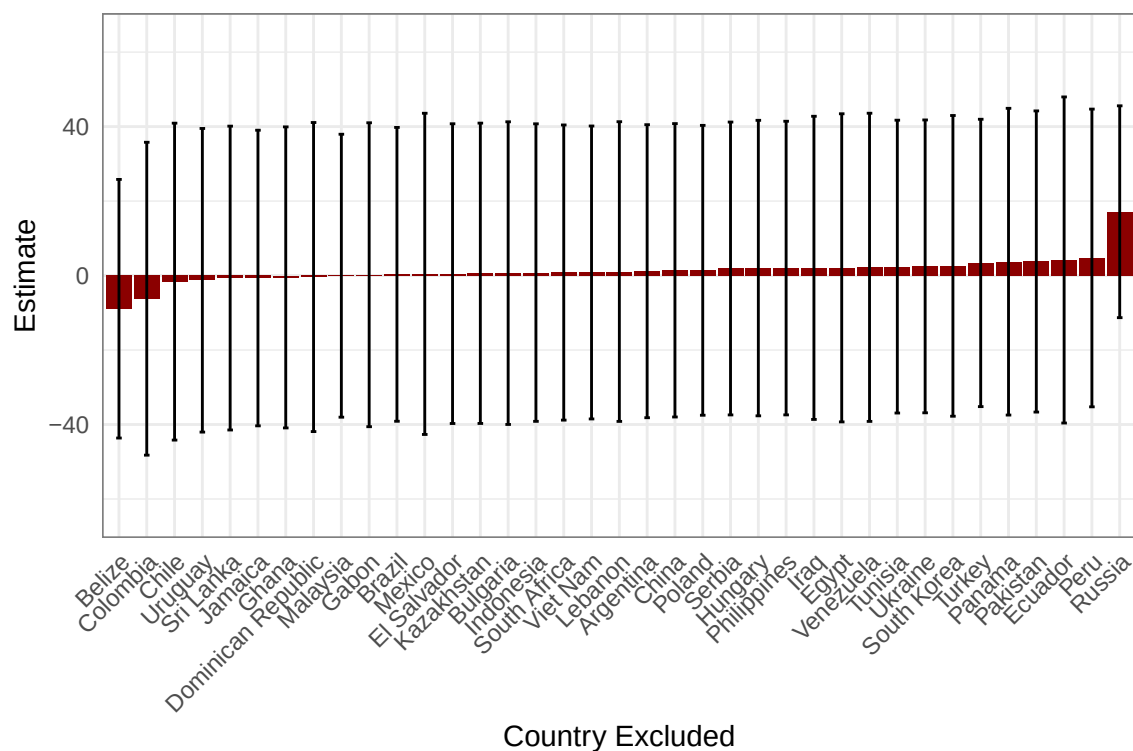
Table 11: Effect on Current Account Dynamics, Excluding Energy Commodities.

	Δ Current Account (% GDP)					
	(1)	(2)	(3)	(4)	(5)	(6)
Import Price Shock	-0.921** (0.420)		-0.858** (0.384)	-0.859** (0.383)	-1.40** (0.572)	-0.877 (0.521)
Export Price Shock		0.381 (0.304)	0.153 (0.266)	0.145 (0.265)	-0.382 (0.380)	-0.168 (0.306)
Observations	728	728	728	728	349	505
R ²	0.20	0.18	0.20	0.22	0.24	0.18
Country fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Extra controls				✓		
Low spreads					✓	
Low debt						✓

Note: This table reports estimated effects on the annual change in the current account balance as a share of GDP. All other controls and sample restrictions follow Table 5. Standard errors are clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

To complement the analysis for import price shocks, Figure 11 presents the leave-one-out estimates for export price shocks, based on the same specification and set of controls as in Figure 2.

Figure 11: Leave-One-Out Estimates: Export Price Shocks.



In addition, Figures 12 and 13 report the corresponding leave-one-out estimates using the full set of controls from the baseline specification, including average maturity and the growth rates of interest and principal debt repayments.

Figure 12: Leave-One-Out Estimates: Import Price Shocks, Including Additional Controls.

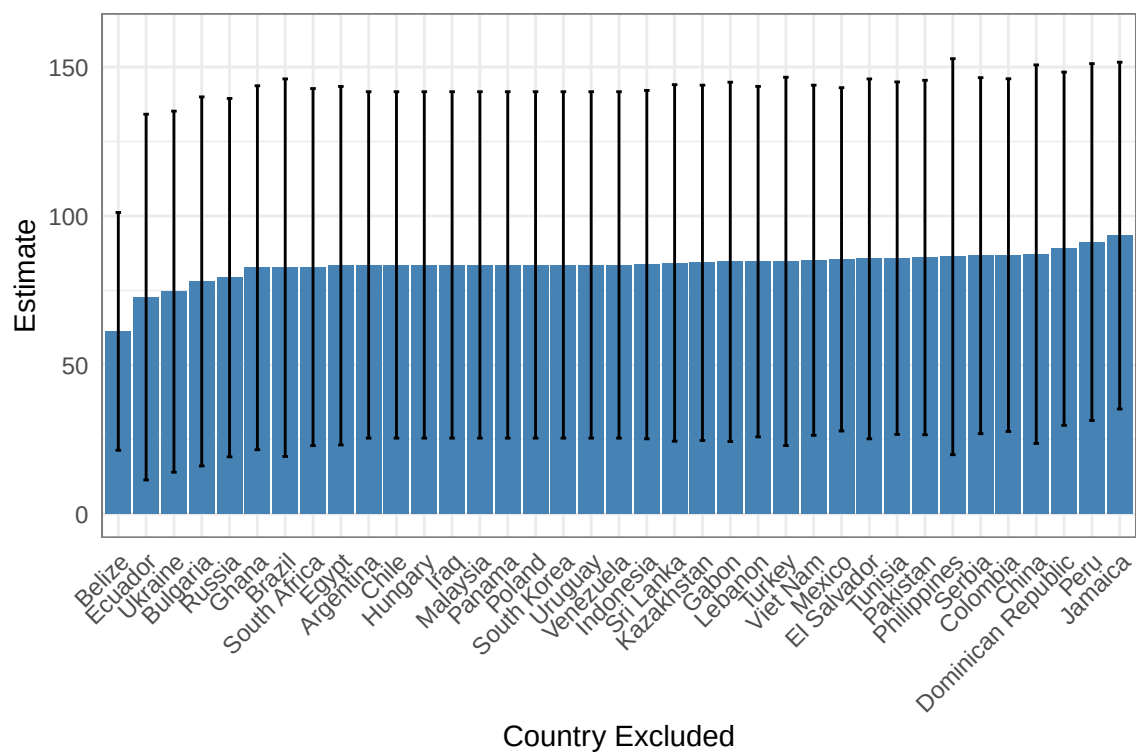


Figure 13: Leave-One-Out Estimates: Export Price Shocks, Including Additional Controls.

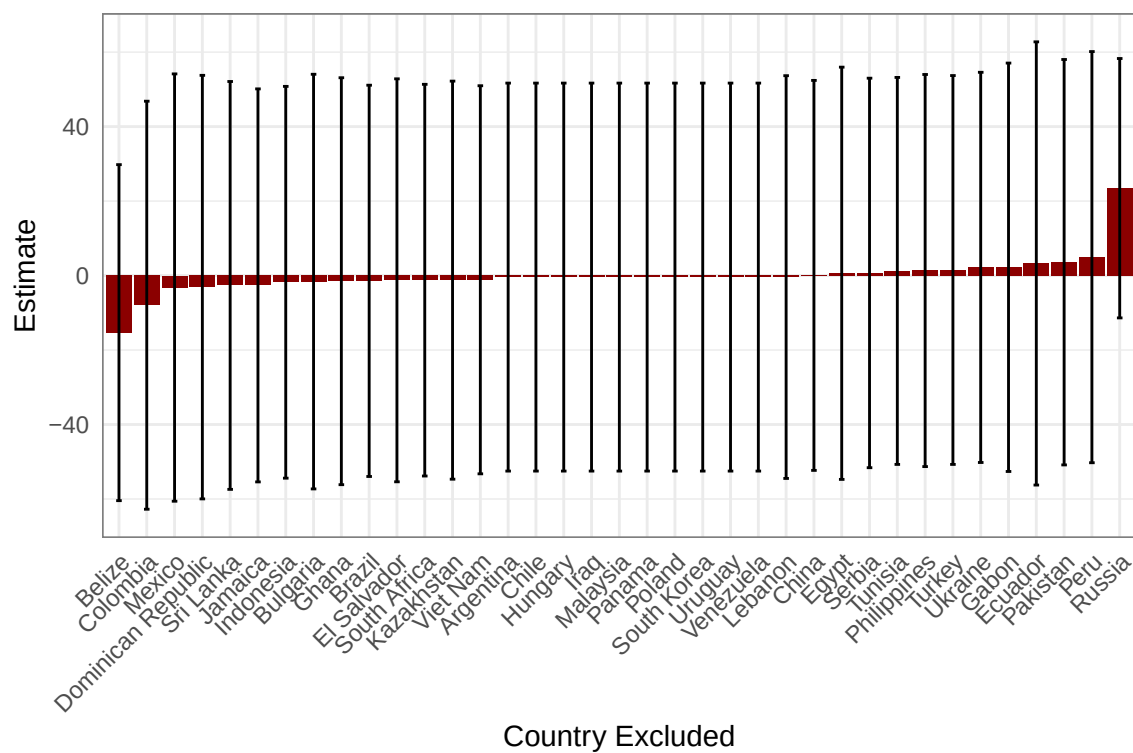
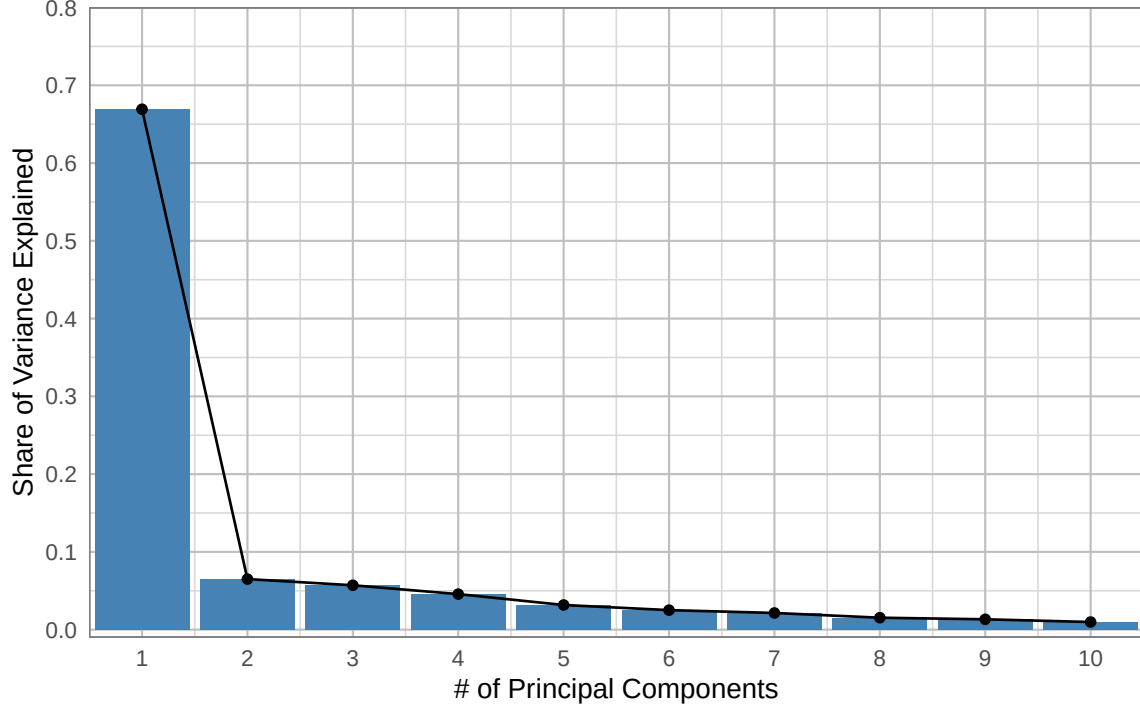


Figure 14 reports the share of variance explained by the first K principal components of log price data for 41 commodities over the period 1991–2023.

Figure 14: Variance Explained by Principal Components.



B Derivations of the Aggregate Environment

This Appendix Section provides the full derivations underlying Section 3.1, including the expenditure minimization problem, the derivation of the price index, and the decomposition of aggregate consumption in factor form.

Price Structure and Factors. The price of each tradable good $i \in [0, 1]$ follows the log-linear factor representation introduced in equation (4):

$$\ln P_t(i) = \mu(i) + a(i)^\top f_t + \varepsilon_t(i),$$

where $f_t \in \mathbb{R}^K$ collects the first K global principal components. The vector of factors evolves according to the stationary VAR(1) process given in equation (5):

$$f_t = \Phi f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u),$$

with eigenvalues of Φ strictly inside the unit circle.

Preferences and Aggregator. Consumption of each good i is denoted $c_t(i)$, with subsistence level $\bar{c}(i)$. The Stone–Geary aggregator is:

$$c_t = \prod_{i \in [0,1]} [c_t(i) - \bar{c}(i)]^{\lambda(i)}, \quad \int_0^1 \lambda(i) di = 1,$$

where $\lambda(i) \geq 0$ is the weight assigned to good i . Taking logs yields:

$$\ln c_t = \int_0^1 \lambda(i) \ln[c_t(i) - \bar{c}(i)] di.$$

Expenditure Minimization and Individual Demands. Given aggregate expenditure E_t , the representative agent minimizes:

$$\int_0^1 P_t(i) c_t(i) di,$$

subject to the aggregator above. The first-order conditions imply:

$$c_t(i) = \bar{c}(i) + \frac{\lambda(i)}{P_t(i)} \left(E_t - \int_0^1 P_t(i') \bar{c}(i') di' \right).$$

where the term in parentheses denotes total expenditure net of subsistence consumption. This expression highlights that, conditional on E_t , additional consumption above subsistence for each good is inversely proportional to its price and weighted by $\lambda(i)$.

Aggregation and the Price Index. Substituting the individual demand functions into the aggregator yields:

$$c_t = \Gamma^c \exp \left(- \int_0^1 \lambda(i) \ln P_t(i) di \right) \left(E_t - \int_0^1 P_t(i) \bar{c}(i) di \right),$$

where $\Gamma^c \equiv \prod_{i \in [0,1]} \lambda(i)^{\lambda(i)}$.

Define the consumption-based price index:

$$P_t^C \equiv \exp \left\{ \int_0^1 \lambda(i) \ln P_t(i) di \right\},$$

such that:

$$c_t = \Gamma^c e^{-\ln P_t^C} \left(E_t - \int_0^1 P_t(i) \bar{c}(i) di \right). \quad (24)$$

Resource Constraint. Define the net endowment $\tilde{e}(i) \equiv e(i) - \bar{c}(i)$. Substituting into the aggregate expenditure identity yields the resource constraint, corresponding to corresponds

to equation (6) in the main text:

$$E_t - \int_0^1 P_t(i) \bar{c}(i) di = D_t + \int_0^1 P_t(i) \tilde{e}(i) di, \quad (25)$$

which corresponds to equation (6) in the main text.

Price Index in Factor Form. Using the factor structure from equation (4), the consumption-based price index inherits the same factor representation:

$$\ln P_t^C = \mu^C + \psi^\top f_t + \varepsilon_t^C, \quad (26)$$

where:

$$\mu^C = \int_0^1 \lambda(i) \mu(i) di, \quad \psi = \int_0^1 \lambda(i) a(i) di, \quad \varepsilon_t^C = \int_0^1 \lambda(i) \varepsilon_t(i) di.$$

The vector ψ captures the sensitivity of the aggregate consumption basket to global price shocks through the price-index channel.

Net-Export Term and Trade Exposure. Let $\bar{P}(i) \equiv e^{\mu(i)}$ denote steady-state prices. A first-order approximation of the trade term around $\bar{P}(i)$ gives:

$$\int_0^1 P_t(i) \tilde{e}(i) di \approx \bar{Z} + \beta^\top f_t + \zeta_t,$$

where:

$$\bar{Z} = \int_0^1 \bar{P}(i) \tilde{e}(i) di, \quad \beta_k = \int_0^1 \bar{P}(i) \tilde{e}(i) a_k(i) di,$$

and the residual term $\zeta_t \equiv \int_0^1 \bar{P}(i) \tilde{e}(i) \varepsilon_t(i) di$ is small when trade exposure is sufficiently diversified.

Aggregate Consumption. Substituting equations (26) and (25) into (24) yields:

$$c_t = \Gamma^c \exp(-\mu^C - \psi^\top f_t - \varepsilon_t^C) (D_t + \bar{Z} + \beta^\top f_t + \zeta_t).$$

Setting the idiosyncratic terms $\varepsilon_t^C = \zeta_t = 0$, a modelling choice equivalent to averaging out residual pricing noise orthogonal to the global factors, we obtain the compact form:

$$c_t = \Gamma^c e^{-\mu^C - \psi^\top f_t} (D_t + \bar{Z} + \beta^\top f_t), \quad (27)$$

which corresponds exactly to equation (7) in the Section 3.1.

C Proofs and Extensions

This appendix collects the analytical proofs of the main theoretical results and presents two extensions. Section C.1 derives the consumption wedge in Proposition 1. Section C.2 characterizes the response of default probabilities and spreads. Finally, Section C.3 establishes sufficient conditions under which these local results extend globally.

C.1 Proof of Proposition 1

Proof. Let $c(f, b)$ be given by:

$$c(f, b) = \Gamma^c \exp(-\mu^C - \psi^\top f) (D(f, b) + \bar{Z} + \beta^\top f),$$

where $D(f, b)$ is the equilibrium net resource flow implied by the optimal debt choice and the bond pricing function. Taking logs and differentiating with respect to factor component f_k yields:

$$\frac{\partial \ln c(f, b)}{\partial f_k} = -\psi_k + \frac{\beta_k + \frac{\partial D(f, b)}{\partial f_k}}{D(f, b) + \bar{Z} + \beta^\top f}. \quad (28)$$

Within a period, the government chooses next-period debt b' after observing f . Write:

$$D(f, b) = q(f, b') b' - b, \quad b' = b'(f, b) \quad (\text{optimal}).$$

We define the consumption wedge as the *impact* elasticity of consumption to the factor, holding the debt policy b' fixed at its current optimum. This is a deliberate modelling choice: it isolates the contemporaneous transmission of a shock through the price-index and terms-of-trade channels, leaving the dynamic response of debt issuance to the companion Corollary 1 and Lemma 1. With b' held fixed,

$$\left. \frac{\partial D(f, b)}{\partial f_k} \right|_{b' \text{ fixed}} = \frac{\partial q(f, b')}{\partial f_k} b',$$

which is itself second order in f around a PCA-normalized reference point $\bar{f} = \mathbf{0}$ (the bond price depends on f_k only through the equilibrium default probability, which is linear in f to first order). Linearizing around $\bar{f} = \mathbf{0}$ therefore gives:

$$\left. \frac{\partial D(f, b)}{\partial f_k} \right|_{b' \text{ fixed}} \approx 0, \quad D(f, b) + \bar{Z} + \beta^\top f \approx D(f, b) + \bar{Z}.$$

Substituting into (28) gives:

$$\frac{\partial \ln c(f, b)}{\partial f_k} = -\psi_k + \frac{\beta_k}{D(f, b) + \bar{Z}} = -\psi_k + \kappa_c(f, b) \beta_k, \quad \kappa_c(f, b) \equiv \frac{1}{D(f, b) + \bar{Z}}.$$

This establishes the expression for the *consumption wedge* $\Delta_k(f, b)$ stated in the proposition.

The sign of $\Delta_k(f, b)$ determines whether a marginal increase in f_k raises or lowers $c(f, b)$: if $\Delta_k > 0$ the terms-of-trade income component $\kappa_c \beta_k$ dominates the price-index drag $-\psi_k$, and vice versa. This establishes part (i).

The leverage term $\kappa_c(f, b) = 1/[D(f, b) + \bar{Z}]$ is larger when $D(f, b) + \bar{Z}$ is smaller (i.e., when the economy has weaker external buffers), which magnifies the contribution of β_k to Δ_k . This establishes part (ii).

For part (iii), apply the envelope theorem to the repayment value:

$$\frac{\partial V^r(f, b)}{\partial f_k} = u'(c) \frac{\partial c(f, b)}{\partial f_k} \Big|_{b' \text{ optimal}} + \beta_d \frac{\partial \mathbb{E}_f[V(f', b')]}{\partial f_k}.$$

With $\Delta_k(f, b) > 0$, the first term is strictly positive, while the second is non-negative when f_k is positively persistent (in the sense of first-order stochastic dominance in the transition kernel) and V inherits monotonicity in f under the wedge condition. It follows that $\partial V^r / \partial f_k > 0$. Applying the same argument to $V^d(f)$, with $c^d = \chi \Gamma^c e^{-\mu^c - \psi^\top f} (\bar{Z} + \beta^\top f)$, yields $\partial V^d / \partial f_k$. We assume that its magnitude is strictly smaller than $\partial V^r / \partial f_k$ at the repayment cutoff, so that the default incentive $V^r - V^d$ is monotone in f_k at $b = b^*(f)$. This corresponds to the standard single-crossing condition in sovereign default models and is verified numerically in all our calibrations. It follows that the default region weakly shrinks in f_k , equilibrium default probabilities weakly decline, and, by the lenders' zero-profit condition, sovereign spreads weakly compress. The opposite holds when $\Delta_k(f, b) < 0$. This establishes part (iii).

The key step relies on the impact derivative evaluated with b' fixed at its current optimum and a first-order expansion around the reference factor state $\bar{f}$. The full dynamic response of b' and prices to f_k is not available in closed form. Corollary 1 and Lemma 1 provide partial characterizations under, respectively, a monotone single-crossing condition on $V^r - V^d$ and a bounded-leverage condition on $D(f, b)$, under which the sign of $\Delta_k(f, b)$ is preserved globally. Both conditions are verified numerically in all calibrations reported in Section 4 and Appendix F. The default heatmaps in Figure 4 and Figures 15-18 illustrate the implied monotone frontiers across all countries in the sample. $\square$

C.2 Global Factors, Default Sets, and Spreads

The next result establishes how the sign of $\Delta_k(f, b)$ translates into equilibrium default probabilities and sovereign spreads extending Proposition 1.

Corollary 1 (Global factors, default sets, and spreads). *Suppose $u'(c) > 0$, $u''(c) \leq 0$; equilibrium repayment is characterized by a single-crossing rule with a cutoff $b^*(f)$ such that the government repays iff $b \geq b^*(f)$; and the default incentive $V^r(f, b) - V^d(f)$ is monotone in f_k at the cutoff, i.e. $\text{sgn}(\partial[V^r - V^d]/\partial f_k) = \text{sgn}(\Delta_k(f, b))$ evaluated at $b = b^*(f)$ (this standard monotone comparative-statics property of sovereign-default problems is verified numerically in all our calibrations). Let bond prices satisfy the zero-profit condition:*

$$q(f, b') = \frac{1 - \delta(f, b')}{1 + r^*}, \quad \delta(f, b') = \int_{\mathcal{D}(b')} \pi(df' | f),$$

where $\mathcal{D}(b') = \{f' : V^r(f', b') < V^d(f')\}$ and $r^* > 0$ is constant.

If, for factor k , the consumption wedge in Proposition 1 satisfies:

$$\Delta_k(f, b) = -\psi_k + \kappa_c(f, b) \beta_k > 0,$$

then a marginal increase in f_k (holding the other components of f fixed) has the following effects:

- (i) **Default set shrinks.** The repayment cutoff weakly decreases, $\partial b^*(f)/\partial f_k \leq 0$; equivalently, the default region $\mathcal{D}(\cdot)$ shrinks in measure. Hence the one-step default probability weakly falls: $\partial \delta(f, b')/\partial f_k \leq 0$ for all b' .
- (ii) **Bond prices rise and spreads compress.** Since $q(f, b')$ is decreasing in $\delta(f, b')$, we have $\partial q(f, b')/\partial f_k \geq 0$ and thus the sovereign spread $s(f, b') \equiv 1/q(f, b') - 1 - r^*$ weakly decreases.

The opposite comparative statics obtain when $\Delta_k(f, b) < 0$.

Proof. Fix (f, b) and perturb factor f_k by a small positive amount. By Proposition 1,

$$\frac{\partial \ln c(f, b)}{\partial f_k} = \Delta_k(f, b).$$

If $\Delta_k(f, b) > 0$, then $\partial c(f, b)/\partial f_k = c(f, b) \Delta_k(f, b) > 0$. By the envelope theorem applied to $V^r(f, b) = \max_{b'} \{u(c(f, b, b')) + \beta_d \mathbb{E}_f[V(f', b')]\}$,

$$\frac{\partial V^r(f, b)}{\partial f_k} = u'(c) \left. \frac{\partial c}{\partial f_k} \right|_{b' \text{ optimal}} + \beta_d \frac{\partial \mathbb{E}_f[V(f', b')]}{\partial f_k}.$$

The first term is strictly positive since $u'(c) > 0$ and $\partial c/\partial f_k > 0$. The second term is non-negative when f_k is positively persistent, that is, when the transition kernel is FOSD-ordered in f_k . A higher f_k then shifts the distribution of f' upward in the FOSD sense, and V is weakly increasing in f' under the wedge condition. Hence, $\partial V^r(f, b)/\partial f_k > 0$.

By the monotone single-crossing property of $V^r - V^d$, the repayment region expands and the default region contracts as f_k increases, so $\delta(f, b')$ weakly decreases. Since $q(f, b') = (1 - \delta(f, b'))/(1 + r^*)$ decreases in δ , bond prices rise and spreads $s = 1/q - (1 + r^*)$ fall. The converse holds when $\Delta_k(f, b) < 0$. $\square$

C.3 Global Comparative Statics Under Bounded Leverage

While Proposition 1 and Corollary 1 characterize local responses to marginal changes in global factors, many relevant episodes, such as commodity price surges or collapses, involve large and persistent shocks. The following lemma provides sufficient conditions under which these qualitative results extend globally, even when $D(f, b)$ and default decisions adjust endogenously to large movements in f_k .

Lemma 1 (Global comparative statics under bounded leverage). *Fix a nonempty set $\mathcal{S} \subset \mathbb{F} \times \mathbb{B}$ of admissible states and suppose the endogenous resource flow satisfies:*

$$D_{\min} \leq D(f, b) \leq D_{\max} \quad \text{for all } (f, b) \in \mathcal{S},$$

with $D_{\min} + \bar{Z} > 0$. Let $\kappa_c(f, b) = 1/[D(f, b) + \bar{Z}]$. Then:

(i) **Exporter dominance.** Suppose $\beta_k > 0$ and

$$\frac{\beta_k}{D_{\max} + \bar{Z}} > \psi_k,$$

so that the terms-of-trade income channel dominates the price-index channel even when the country sits on the largest admissible external buffer. Then, for all $(f, b) \in \mathcal{S}$, the map $f_k \mapsto c(f, b)$ is strictly increasing. Moreover, $V^r(f, b)$ is strictly increasing in f_k , the default set shrinks (in the sense of set inclusion), and bond prices $q(f, b')$ weakly increase.

(ii) **Importer dominance.** Suppose $\beta_k < 0$ and

$$\frac{|\beta_k|}{D_{\min} + \bar{Z}} > \psi_k,$$

so that the (negative) terms-of-trade channel outweighs any non-negative price-index effect even at the tightest admissible buffer. Then for all $(f, b) \in \mathcal{S}$ the map $f_k \mapsto c(f, b)$ is strictly decreasing. Consequently, $V^r(f, b)$ weakly decreases in f_k , the default set expands, and $q(f, b')$ weakly falls.

Proof. Fix $(f, b) \in \mathcal{S}$ and consider a change in factor f_k . From the consumption expression,

$$c(f, b) = \Gamma^c e^{-\mu^C - \psi^\top f} (D(f, b) + \bar{Z} + \beta^\top f),$$

the impact derivative (holding D fixed) is:

$$\left. \frac{\partial \ln c(f, b)}{\partial f_k} \right|_D = -\psi_k + \frac{\beta_k}{D(f, b) + \bar{Z} + \beta^\top f}.$$

Evaluating at $\beta^\top f = 0$, as in Proposition 1, yields:

$$\left. \frac{\partial \ln c(f, b)}{\partial f_k} \right|_D = -\psi_k + \frac{\beta_k}{D(f, b) + \bar{Z}}.$$

Since $D(f, b) \in [D_{\min}, D_{\max}]$ and $D_{\min} + \bar{Z} > 0$, the term $1/(D(f, b) + \bar{Z})$ is positive and bounded.

Exporter dominance. If $\beta_k > 0$ and $\beta_k/(D_{\max} + \bar{Z}) > \psi_k$, then

$$\left. \frac{\partial \ln c(f, b)}{\partial f_k} \right|_D > 0 \quad \text{for all } (f, b) \in \mathcal{S}.$$

Hence $c(f, b)$ is strictly increasing in f_k globally. By the envelope theorem, $\partial V^r(f, b)/\partial f_k > 0$. Under the monotone single-crossing property of $V^r - V^d$, the default set shrinks and equilibrium default probabilities weakly decrease. By the lenders' zero-profit condition, bond prices $q(f, b')$ weakly increase.

Importer dominance. If $\beta_k < 0$ and $|\beta_k|/(D_{\min} + \bar{Z}) > \psi_k$, then

$$\left. \frac{\partial \ln c(f, b)}{\partial f_k} \right|_D < 0 \quad \text{for all } (f, b) \in \mathcal{S}.$$

Hence $c(f, b)$ is strictly decreasing in f_k . The same argument implies $\partial V^r(f, b)/\partial f_k < 0$, the default set expands, and bond prices $q(f, b')$ weakly decrease.

These bounds are uniform on $\mathcal{S}$, and the endogenous response of $D(f, b)$ does not overturn the sign of the impact derivative. This establishes the result. $\square$

D Estimation and Solution Algorithm

This Appendix Section describes the empirical construction of the exposure parameters $(\psi, \beta, \bar{Z}, \mu^C)$ and the numerical procedure used to compute the equilibrium. The structure mirrors the calibration framework in Section 4.1. We first estimate the global factor structure and the exposure parameters from the data and then, conditional on these inputs, solve the government's optimization problem and the associated market-clearing bond price schedule.

Link to PCA Facts. The factorization:

$$\ln P_t(i) = \mu(i) + a(i)^\top f_t + \varepsilon_t(i)$$

mirrors the empirical evidence presented in Section 2. On the quarterly detrended log-price panel that enters the solver, the first principal component (PC1) explains more than 60% of the total variance of international prices, and the first six principal components account for about 90%, consistent with the corresponding empirical figure in Section 4.1.

The persistence of global shocks is captured by the process:

$$f_t = \Phi f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u),$$

with eigenvalues of Φ strictly inside the unit circle.

Idiosyncratic components $\varepsilon_t(i)$ are near white noise, consistent with the empirical autocorrelations after purging principal components.

Two Exposure Vectors. Aggregate sensitivity to global shocks is governed by two exposure vectors:

$$(i) \text{ Price-index channel: } -\psi^\top f_t, \quad (ii) \text{ Terms-of-trade channel: } +\beta^\top f_t.$$

Export concentration in goods with high factor loadings $a_k(i)$ increases β_k , while concentration of consumption weights $\lambda(i)$ on goods with high $a_k(i)$ increases ψ_k .

Calibration and Estimation Steps.

1. **Principal Component Estimation.** Let N denote the number of commodities (in our case, $N = 41$). Construct the $T \times N$ matrix of demeaned log prices:

$$X = [\ln P_{t,i} - \ln \bar{P}_i]_{t=1,\dots,T; i=1,\dots,N}.$$

We perform PCA on X (covariance PCA) to obtain the factor representation:

$$X \approx F A^\top, \quad F \in \mathbb{R}^{T \times K}, \quad A \in \mathbb{R}^{N \times K},$$

where the i th row of A , denoted $a(i)^\top$, contains the loadings for commodity i , so that

$$\ln P_{t,i} \approx \ln \bar{P}_i + a(i)^\top f_t.$$

Here $f_t \in \mathbb{R}^K$ is the vector of factor realizations at time t , and $\bar{P}_i \equiv \exp(\ln \bar{P}_i)$ is the baseline price. The factors are normalized such that $\mathbb{E}[f_t] = 0$ and $\mathbb{E}[f_t f_t^\top] = I_K$.

2. **Factor Dynamics.** We estimate the autoregressive process:

$$f_t = \Phi f_{t-1} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma_u),$$

by OLS, and discretize it into a finite Markov chain $\{f^j, P_f(j, j')\}$ using the quadrature method of [Tauchen & Hussey \(1991\)](#).

3. **Inputs from Consumption and Trade Data.** We construct: (i) consumption weights $\lambda = (\lambda_1, \dots, \lambda_N)$ with $\sum_i \lambda_i = 1$; (ii) net endowments $\tilde{e} = (\tilde{e}_1, \dots, \tilde{e}_N)$, where $\tilde{e}(i) = e(i) - \bar{c}(i)$; and (iii) baseline prices $\bar{P}_i = e^{\mu(i)}$. These objects map the observed consumption and trade structure into model inputs.

4. **Computation of Exposure Parameters.** We compute:

$$\begin{aligned} \psi &= A^\top \lambda, & \beta &= A^\top (\bar{P} \odot s), \\ \bar{Z} &= \sum_{i=1}^N \bar{P}_i s_i, & \mu^C &= \sum_{i=1}^N \lambda_i \mu(i), \end{aligned}$$

where $\odot$ denotes elementwise multiplication. The pair (ψ, β) captures the price-index and terms-of-trade channels, while $(\bar{Z}, \mu^C)$ pin down the steady-state trade balance and average consumption price index.

5. **Country-Specific Calibration.** For each of the 37 countries, we observe production values V_{ij}^P , exports X_{ij} , and imports M_{ij} for each commodity j , and compute the country-specific exposure pair $(\psi_i, \tilde{\beta}_i)$ as described in [Appendix E](#). Countries with $\tilde{\beta}_i > 0$ are net exporters of high-loading commodities and benefit from global price expansions, whereas those with $\tilde{\beta}_i < 0$ are net importers. The consumption exposure ψ_i reflects the composition of domestic consumption.

6. **Solution of the Model.** Given $(\psi, \beta, \bar{Z}, \mu^C)$ and the calibrated parameters, we solve for equilibrium recursively:

- (a) Initialize a bond price schedule $q^{(0)}(f, b') = 1/(1+r^*)$ and value functions $V^{r,(0)}(f, b)$ and $V^{d,(0)}(f)$.
- (b) For each state (f, b) , compute the repayment value:

$$V^r(f, b) = \max_{b'} \left\{ u(c^r(f, b, b')) + \beta_d \sum_{f'} P_f(f, f') V(f', b') \right\},$$

where

$$c^r(f, b, b') = \Gamma^c e^{-\mu^C - \psi^\top f} [b - q(f, b')b' + \bar{Z} + \beta^\top f].$$

The default value is:

$$V^d(f) = u(c^d(f)) + \beta_d \sum_{f'} P_f(f, f') [\theta V(f', 0) + (1 - \theta) V^d(f')],$$

with

$$c^d(f) = \chi \Gamma^c e^{-\mu^c - \psi^\top f} (\bar{Z} + \beta^\top f).$$

- (c) Update $V(f, b) = \max\{V^r(f, b), V^d(f)\}$ and define the default region $\mathcal{D}(b) = \{f : V^d(f) > V^r(f, b)\}$.
- (d) Compute default probabilities $\delta(f, b') = \sum_{f' \in \mathcal{D}(b')} P_f(f, f')$ and update bond prices:

$$q^{(1)}(f, b') = \frac{1 - \delta(f, b')}{1 + r^*}.$$

- (e) Iterate until convergence of $q(f, b')$ and the value functions.

At convergence, equilibrium policies, prices, and default probabilities jointly satisfy optimality and market clearing.

Overall, Steps (1)–(5) discipline the model by estimating exposure parameters from international price data, while Step (6) solves the model numerically via value function iteration. The equilibrium highlights how global price shocks interact with trade exposure to determine consumption, debt, and default risk. Throughout, we set $K = 1$.

E Cross-Country Calibration of (ψ_i, β_i)

This appendix details the construction of the country-specific exposure parameters (ψ_i, β_i) used in the quantitative analysis. The objective is to map the model's theoretical objects - net endowments, baseline prices, and consumption weights - into observable counterparts for each of the 37 countries and the $J = 41$ IMF primary commodities.

The procedure mirrors the structure of the model. We first describe the data and commodity coverage, then present the construction of the exposure parameters, and finally report the resulting cross-country estimates and classifications.

E.1 Data Sources and Commodity Coverage

The calibration combines three data sources at the commodity level. Log prices are taken from the IMF Primary Commodity Prices Database. The time-average log price defines the baseline price $\bar{P}(j) = \exp(\mu(j))$, and the same panel is used to estimate the PCA loadings $A(j)$. Using a single source ensures consistency between price levels and factor

loadings. Export and import values (USD) are obtained from UN Comtrade at the SITC Revision 1 level. We construct average annual exports X_{ij} and imports M_{ij} over 1995–2019, excluding major crisis episodes. Values are measured at international prices and are therefore directly comparable to IMF price data. Production quantities Q_{ij}^P are sourced from FAOSTAT (agriculture and forestry), the U.S. EIA (energy), and USGS/UN Data (metals and minerals). These quantities are converted into values using IMF prices:

$$V_{ij}^P = Q_{ij}^P \bar{P}(j).$$

The resulting dataset covers 41 commodities spanning energy, metals, agriculture, and livestock. A detailed crosswalk between SITC codes, FAO items, and production sources is reported in Table 12. Unit conversions (e.g. cubic meters to tonnes, Bcf to tonnes) ensure comparability across sources.

Table 12: Commodity Crosswalk – SITC, FAO, and Production Sources.

Commodity	SITC Rev. 1	FAO Item Code	Production Source
Cotton	263, 652, 6513, 6514	767, 328	FAO
Hard logs	2423	1863, 1867, 1657, 1670	FAO
Soft logs	2422	1862, 1866, 1651	FAO
Hard sawnwood	2433	1633, 1644	FAO
Soft sawnwood	2432	1632, 1643	FAO
Hides	2111, 2119	957, 919, 1025	FAO
Natural rubber	2311	836, 839	FAO
Wool	262, 6512	987	FAO
Coal	32	-	U.S. EIA
Natural gas	3411	-	U.S. EIA
Crude oil	331	-	U.S. EIA
Bananas	0513	486, 489	FAO
Barley	043	44	FAO
Beef	0111	947, 867	FAO
Chicken	0114	1058, 1080	FAO
Cocoa	072	661	FAO
Coffee	071	656	FAO
Corn	044	56	FAO
Fish	0311, 0312, 03201	1176	FAO
Groundnuts	2211	242	FAO

(continued)

Commodity	SITC Rev. 1	FAO Item Code	Production Source
Lamb	0112	977, 1017	FAO
Olive oil	4215	261	FAO
Oranges	05111	490	FAO
Palm oil	4222	257	FAO
Pork	0113	1035	FAO
Rapeseed oil	4217	271	FAO
Rice	042	27	FAO
Soybean oil	4212	237	FAO
Soybeans	2214	236	FAO
Sugar	0611, 0612	162	FAO
Sunflower seed oil	4216	281, 268	FAO
Tea	0741	667	FAO
Wheat	0410	15	FAO
Aluminum	684	-	USGS
Copper	2831, 682	-	USGS
Iron ores	2813	-	USGS
Lead	2834, 685	-	USGS
Nickel	28321, 683	-	USGS
Tin	2836, 687	-	USGS
Uranium	286, 688	-	U.N. Data
Zinc	2835, 686	-	USGS

Note: SITC Rev. 1 codes are used to extract trade flows from UN Comtrade. FAO item codes correspond to FAOSTAT crop and livestock production data. Production sources labeled USGS refer to the U.S. Geological Survey Minerals Yearbook; U.S. EIA refers to the U.S. Energy Information Administration. A dash (-) indicates that FAO or an alternative source is not the primary production provider for that category.

E.2 Construction of Exposure Parameters

We now map the data into the model objects. First, the empirical counterpart of $\bar{P}(i)\tilde{e}(i)$ is the value of net exports:

$$NX_{ij} = X_{ij} - M_{ij}.$$

Domestic absorption is:

$$C_{ij} = \max\{V_{ij}^P - X_{ij} + M_{ij}, 0\},$$

where the truncation accounts for re-export activity, V_{ij}^P is the average value of domestic production in USD for a specific country i and a specific commodity j , X_{ij} is the average value of exports in USD, and M_{ij} is the average value of imports in USD. Both imports and exports data are computed from *UN COMTRADE*.

Replacing integrals by sums yields:

$$\bar{Z}_i = \sum_j NX_{ij}, \quad \beta_i = \sum_j NX_{ij} a(j), \quad \psi_i = \sum_j \lambda_{ij} a(j),$$

with $\lambda_{ij} = C_{ij} / \sum_k C_{ik}$. These correspond directly to the model definitions of trade income, income exposure, and price-index exposure in equation (7).

The parameter β_i captures the sensitivity of trade income to global price movements, while ψ_i captures the response of the consumption price index. Their difference determines the sign of the consumption wedge and therefore the response of default incentives.

To ensure comparability across countries, we define the normalized exposure:

$$\tilde{\beta}_i = \frac{\beta_i}{|\bar{Z}_i|}.$$

This preserves the economic sign of β_i and avoids mechanical sign reversals for net importers. The model normalization sets $\bar{Z} \equiv 1$ and $\bar{\kappa} = 1$, so that $(\psi_i, \tilde{\beta}_i)$ can be directly used in the solver.

Implementation The construction to map the observed trade and production structure of each country into the two key parameters governing the model proceeds as follows:

1. Assemble X_{ij} , M_{ij} , and Q_{ij}^P for each country and commodity.
2. Convert production to values using IMF prices.
3. Compute NX_{ij} and C_{ij} .
4. Construct $\bar{Z}_i$, β_i , ψ_i using factor loadings $a(j)$.
5. Normalize β_i to obtain $\tilde{\beta}_i$.

Outliers and Edge Cases Two features of the data are worth noting. First, ψ_i varies relatively little across countries, reflecting the fact that all countries consume energy and metals, which carry high factor loadings. Cross-country heterogeneity therefore primarily

arises from differences in $\tilde{\beta}_i$. Second, countries with $\bar{Z}_i \approx 0$ (notably Ukraine and Belize) exhibit unstable normalized exposures. For these countries, small differences between exports and imports lead to large variation in $\tilde{\beta}_i$, so results should be interpreted with caution.

E.3 Country-Level Estimates

Table 13 reports the estimated parameters $(\psi_i, \tilde{\beta}_i)$ and the implied wedge:

$$\hat{\Delta}_i = -\psi_i + \tilde{\beta}_i.$$

A positive wedge identifies exporter-type countries, for which the income channel dominates and global price increases reduce default risk. A negative wedge identifies importer-type countries, for which the price-index channel dominates and global price increases raise default risk.

The estimates display substantial heterogeneity. While ψ_i lies in a narrow range (approximately 0.14—0.18), $\tilde{\beta}_i$ varies widely, reflecting differences in trade specialization. As a result, the sign of $\hat{\Delta}_i$ is primarily driven by trade structure.

Exporter-type countries include major commodity producers such as Argentina, Chile, Mexico, and South Africa. Importer-type countries include manufacturing-oriented economies such as China, Poland, and Hungary. Several countries lie close to the threshold, reflecting near-offsetting income and price-index channels.

These calibrated parameters provide the empirical foundation for the cross-country comparative statics in the main text. The sign of $\hat{\Delta}_i$ determines whether global commodity shocks amplify or mitigate default risk, consistent with Proposition 1.

Table 13: Country-Level Exposure Parameters.

Country	ψ_i	$\tilde{\beta}_i$	$\hat{\Delta}_i$	Type
Argentina	0.1673	0.1714	0.0041	Exporter
Belize	0.1682	0.1688	0.0006	Exporter
Brazil	0.1670	0.1705	0.0035	Exporter
Bulgaria	0.1761	0.1609	-0.0152	Importer
Chile	0.1552	0.1794	0.0242	Exporter
China	0.1725	0.1711	-0.0015	Importer
Colombia	0.1716	0.1701	-0.0015	Importer
Dominican Republic	0.1536	0.1601	0.0064	Exporter
Ecuador	0.1703	0.1679	-0.0024	Importer

(continued)

Country	ψ_i	$\tilde{\beta}_i$	$\hat{\Delta}_i$	Type
Egypt	0.1694	0.1648	-0.0046	Importer
El Salvador	0.1546	0.1669	0.0123	Exporter
Gabon	0.1708	0.1685	-0.0023	Importer
Ghana	0.1685	0.1663	-0.0022	Importer
Hungary	0.1703	0.1654	-0.0048	Importer
Indonesia	0.1737	0.1716	-0.0021	Importer
Iraq	0.1708	0.1717	0.0009	Exporter
Jamaica	0.1568	0.1627	0.0059	Exporter
Kazakhstan	0.1720	0.1712	-0.0007	Importer
Lebanon	0.1428	0.1611	0.0183	Exporter
Malaysia	0.1704	0.1603	-0.0101	Importer
Mexico	0.1701	0.1900	0.0198	Exporter
Pakistan	0.1468	0.2056	0.0588	Exporter
Panama	0.1528	0.0910	-0.0617	Importer
Peru	0.1659	0.1792	0.0133	Exporter
Philippines	0.1568	0.1671	0.0103	Exporter
Poland	0.1753	0.1620	-0.0133	Importer
Russia	0.1715	0.1668	-0.0047	Importer
Serbia	0.1752	0.1599	-0.0153	Importer
South Africa	0.1765	0.2297	0.0533	Exporter
South Korea	0.1669	0.1664	-0.0005	Importer
Sri Lanka	0.1424	0.1804	0.0380	Exporter
Tunisia	0.1673	0.1756	0.0082	Exporter
Turkey	0.1681	0.1685	0.0004	Exporter
Ukraine	0.1738	-0.0500	-0.2238	Importer
Uruguay	0.1434	0.1472	0.0038	Exporter
Venezuela	0.1707	0.1710	0.0003	Exporter

Viet Nam	0.1682	0.1539	-0.0143	Importer
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Note: ψ_i is the consumption-expenditure-weighted average of factor loadings $A(j)$. $\tilde{\beta}_i = \beta_i / \bar{Z}_i$ is the normalized net-export-weighted average of factor loadings, where $\beta_i = \sum_j NX_{ij} \cdot A(j)$ and $\bar{Z}_i = \sum_j NX_{ij}$. $\hat{\Delta}_i = -\psi_i + \bar{\kappa}^c \tilde{\beta}_i$ with $\bar{\kappa}^c = 1$ (leverage evaluated at $D_0 = 0$). Reference period: 1995–2019. Factor loadings $A(j)$ are PCA estimates from the IMF commodity price panel. A country is classified as Exporter-type if $\hat{\Delta}_i > 0$ and Importer-type if $\hat{\Delta}_i < 0$.

F Cross-Country Quantitative Results

This Appendix Section extends the quantitative analysis of Section 4.2 to the full sample of 37 countries. For each country, we solve the model using its country-specific exposure parameters $(\psi_i, \tilde{\beta}_i)$ and simulate 500 histories of 10,000 periods. Countries are then classified according to the sign of their response to a positive innovation in the global commodity factor, measured by $\text{Cov}(\text{Annual Spread}, F)$.

A country is *exporter-type* when its probability of default decreases following a rise in global commodity prices. This occurs when the terms-of-trade income channel dominates the price-index channel: the country holds large net-export positions in commodities with high exposure to the global factor, so the associated increase in trade income more than offsets the higher cost of the consumption basket. Real consumption rises, the marginal utility of consumption falls, and repayment becomes easier.

A country is *importer-type* when its probability of default increases following a rise in global commodity prices. In this case, the price-index channel dominates: the country net-imports the commodities most exposed to the global factor, so higher global prices both worsen its terms of trade and raise the domestic price index. The two channels reinforce each other, compressing real consumption, increasing marginal utility, and strengthening default incentives.

This classification corresponds directly to Proposition 1. The relative strength of the terms-of-trade income channel and the price-index channel determines the slope of the default frontier with respect to the global factor, and therefore whether sovereign spreads are countercyclical or procyclical.

The results are reported in three blocks: simulated moments (Table 14), default sets and bond price schedules (Figures 15–22), and dynamics around default events (Figure 23). Within each block, exporter-type countries are presented first, followed by importer-type countries. Within each group, countries are ordered by the magnitude of their exposure to the global factor.

F.1 Simulation Moments: Full Sample

Table 14 reports the counterpart to Table 7 for the full sample of 37 countries, grouped by wedge type. The set of variables and the column structure are identical to those in Table 7: Annual (%), Debt, Consumption, and the Global Factor, each reported with their mean, standard deviation, and covariance with the global factor.

Table 14: Simulation Moments Across All 37 Sample Countries.

Country	Annual Spread (%)			Debt			Consumption		
	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
<i>Exporter-type</i> (20 countries)									
South Africa	2.69	19.83	-4.35	0.168	0.120	0.113	0.966	0.090	0.076
Chile	0.82	4.64	-1.35	0.218	0.113	0.108	0.979	0.048	0.035
Mexico	1.15	8.76	-1.97	0.209	0.116	0.111	0.976	0.050	0.033
Peru	0.84	4.60	-1.34	0.223	0.115	0.110	0.978	0.042	0.025
Argentina	0.70	3.78	-1.11	0.231	0.113	0.109	0.980	0.035	0.014
Brazil	0.70	3.78	-1.12	0.231	0.113	0.109	0.980	0.035	0.014
Uruguay	0.39	2.00	-0.68	0.258	0.102	0.099	0.985	0.027	0.009
Belize	0.72	6.74	-1.48	0.233	0.112	0.107	0.980	0.034	0.011
Iraq	0.71	3.79	-1.13	0.232	0.113	0.109	0.980	0.035	0.011
Venezuela	0.70	3.78	-1.11	0.232	0.113	0.109	0.980	0.034	0.011
Kazakhstan	0.70	3.78	-1.11	0.232	0.113	0.109	0.980	0.034	0.009
Colombia	0.69	3.79	-1.11	0.231	0.113	0.109	0.980	0.034	0.009
Indonesia	0.71	3.79	-1.13	0.232	0.113	0.109	0.980	0.034	0.008
Ecuador	0.68	4.58	-1.12	0.233	0.112	0.107	0.981	0.033	0.008
Gabon	0.68	4.58	-1.12	0.233	0.112	0.107	0.981	0.033	0.007
Russia	0.65	4.42	-1.11	0.235	0.110	0.106	0.981	0.032	0.005
Ghana	0.59	4.89	-1.20	0.242	0.111	0.107	0.982	0.030	0.003
Viet Nam	0.57	3.45	-0.94	0.243	0.110	0.106	0.982	0.030	0.001
Malaysia	0.56	3.44	-0.93	0.243	0.110	0.106	0.982	0.030	-0.002
Ukraine	0.05	0.08	-0.01	0.381	0.042	0.039	1.006	0.147	-0.149
<i>Importer-type</i> (17 countries)									
Panama	0.03	0.06	0.02	0.379	0.026	-0.021	1.019	0.229	-0.231

(continued)

Country	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
Lebanon	0.41	1.98	0.73	0.165	0.122	0.000	1.033	0.320	-0.323
Dominican Republic	0.43	7.13	2.36	0.158	0.134	0.010	1.036	0.332	-0.334
Jamaica	0.41	2.60	0.82	0.146	0.141	0.020	1.038	0.338	-0.340
El Salvador	0.45	3.96	1.10	0.141	0.144	0.022	1.038	0.339	-0.341
Sri Lanka	0.64	3.81	1.20	0.116	0.152	0.028	1.036	0.339	-0.342
Philippines	0.46	3.28	0.94	0.135	0.149	0.025	1.038	0.343	-0.345
Egypt	0.45	2.81	0.89	0.132	0.159	0.035	1.042	0.354	-0.355
South Korea	0.43	2.45	0.81	0.126	0.160	0.036	1.042	0.354	-0.355
Serbia	0.44	7.57	2.46	0.138	0.159	0.036	1.044	0.357	-0.358
Hungary	0.43	3.88	1.10	0.125	0.163	0.039	1.043	0.357	-0.358
Turkey	0.48	3.47	0.98	0.121	0.164	0.041	1.043	0.356	-0.358
Poland	0.38	2.00	0.73	0.133	0.162	0.038	1.044	0.358	-0.360
Bulgaria	0.38	2.72	0.93	0.133	0.162	0.038	1.044	0.359	-0.360
Tunisia	0.52	3.15	1.00	0.110	0.170	0.046	1.043	0.359	-0.361
China	0.50	5.23	1.30	0.107	0.174	0.050	1.045	0.364	-0.366
Pakistan	1.51	19.19	3.21	0.029	0.204	0.087	1.040	0.371	-0.373

Note: The global factor itself has Mean ≈ 0.002 and Std ≈ 1.018 across all 37 countries.

Note: “Cov.” denotes the sample covariance with the global factor. The global factor itself has a mean of ≈ 0.002 and a standard deviation of ≈ 1.018 across all 37 countries.

Cross-country patterns. The cross-country results align closely with the mechanism developed in Section 4.2. Four regularities emerge.

(i) *Sharp separation in spread cyclicalities.* The sign of spread cyclicalities is perfectly aligned with the model classification. For all exporter-type countries, $\text{Cov}(\text{Spread}, F)$ is negative, while for all importer-type countries it is positive. The cross-sectional range is wide, from -4.35 (South Africa) to $+3.21$ (Pakistan), with no country crossing the zero threshold. This clean separation is the direct implication of the sign of the wedge Δ_i , which governs the response of default incentives to global shocks.

(ii) *Strong amplification of consumption risk in importer-type economies.* Consump-

tion volatility is substantially higher in importer-type countries, with an average standard deviation of 0.343 compared to 0.043 in exporter-type countries. The covariance $\text{Cov}(\text{Consumption}, F)$ also switches sign, being close to zero or slightly positive in exporter-type economies and strongly negative in importer-type economies. This reflects the interaction of the two channels: in importer-type countries, the price-index and terms-of-trade effects reinforce each other, whereas in exporter-type countries they partially offset.

(iii) Spread dynamics are driven by tail realizations. Across countries, spread volatility exceeds mean spreads by a large margin, typically by a factor of five or more. This pattern indicates that sovereign risk is dominated by tail realizations of the global factor rather than average conditions. Within the importer group, Pakistan exhibits both the highest mean spread (1.52%) and the largest volatility (19.2%), followed by Sri Lanka and China. Among exporters, South Africa displays both high average spreads and extreme cyclical, reflecting a large positive $\tilde{\beta}_i$ that steepens the default frontier and amplifies responses to global shocks.

(iv) Limited dispersion in borrowing despite large differences in spreads. Debt-to-consumption ratios lie in a relatively narrow range, between 0.14 and 0.26 across countries, and $\text{Cov}(\text{Debt}, F)$ is positive in both groups. Global expansions relax borrowing constraints in exporter-type economies and induce higher borrowing in importer-type economies prior to the tightening associated with higher prices. As a result, cross-country differences in spreads primarily reflect variation in default incentives along the frontier rather than differences in borrowing levels.

F.2 Default Sets for the 37 Countries

Figures 15-18 report the default sets $D(b, f)$ for all countries in the sample, grouped by wedge type and ordered by the magnitude of their exposure to the global factor. Each heatmap shows the combinations of debt b and global factor f for which default is optimal. The slope of the frontier is governed by the sign of the wedge Δ_i , while its steepness reflects the magnitude of exposure to the global factor.

The shape of the default frontier reflects the relative strength of the two channels. In exporter-type countries, the frontier slopes downward in f : default occurs at low realizations of the global factor, where adverse terms-of-trade shocks reduce real income and increase default incentives. In importer-type countries, the frontier slopes upward: default occurs at high realizations of the global factor, where increases in the price of the consumption basket compress real income.

The steepness of the frontier increases with exposure to the global factor. Countries with larger absolute exposure display sharper transitions between repayment and default regions,

consistent with a stronger sensitivity of consumption and default incentives to global shocks. This pattern is particularly visible for highly exposed economies such as Mexico and Chile among exporters, and China and Poland among importers.

Figure 15: Default Sets – Exporter-type Countries, Part 1.

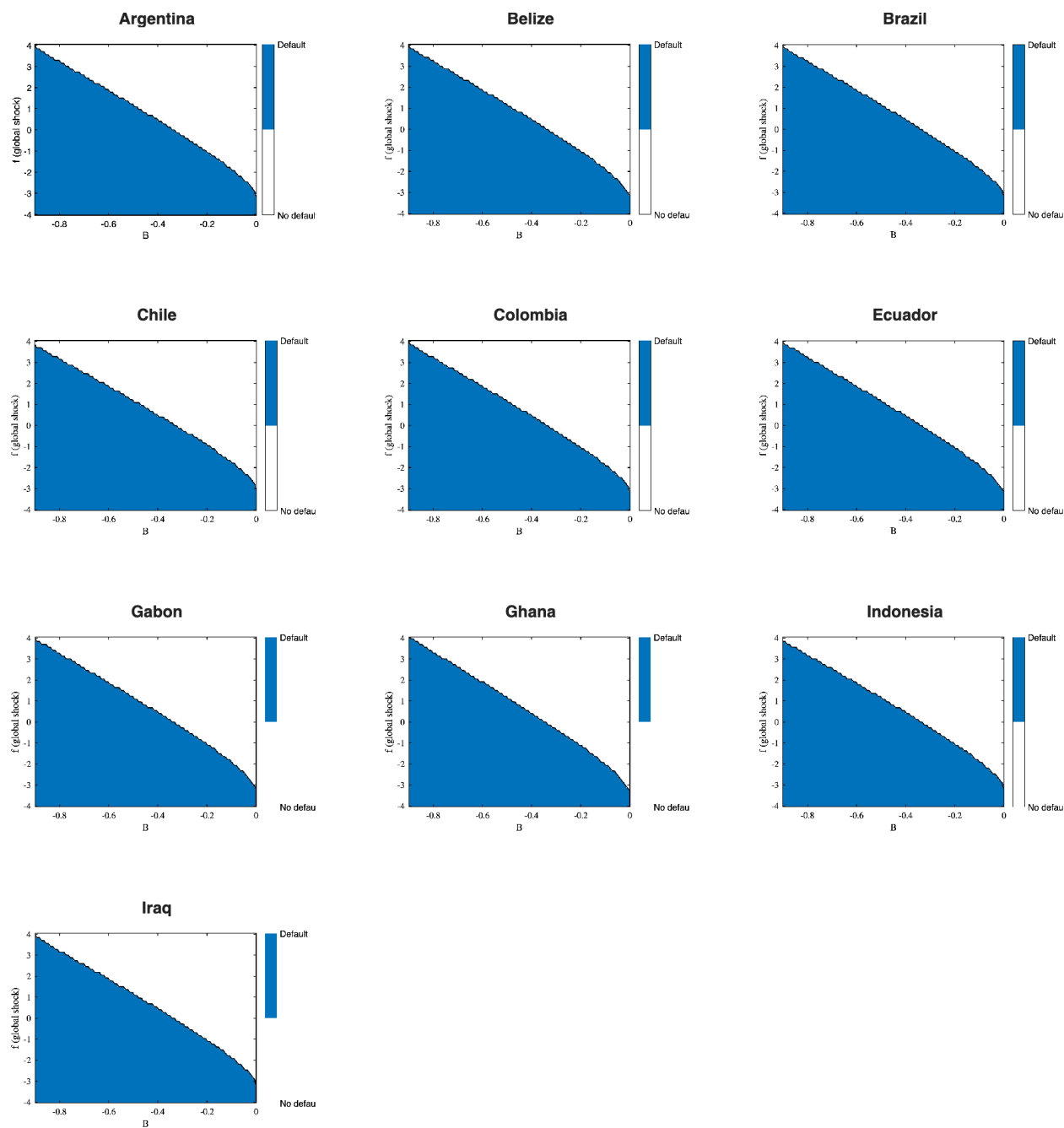


Figure 16: Default Sets – Exporter-type Countries, Part 2.

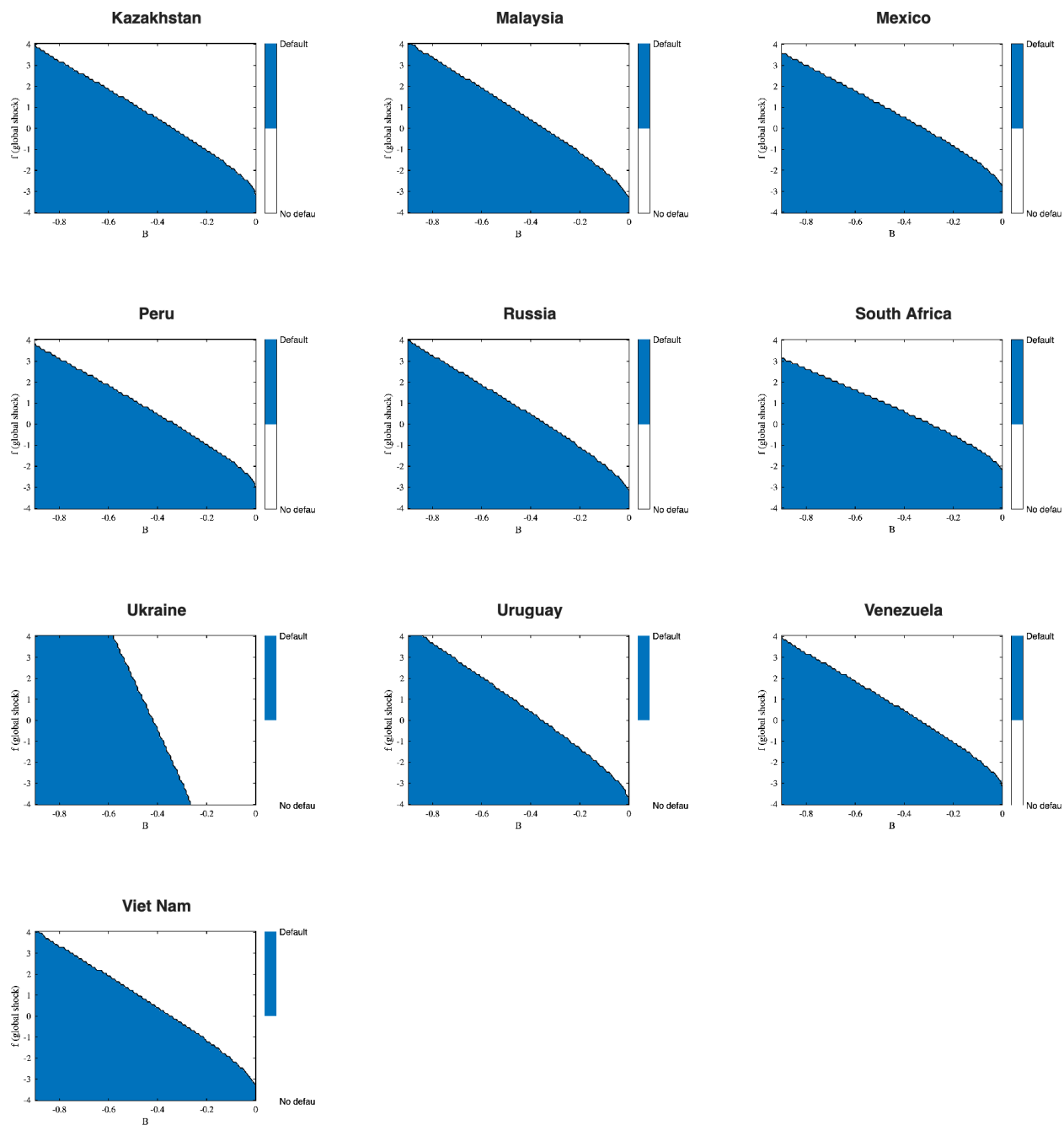


Figure 17: Default Sets – Importer-type Countries, Part 1.

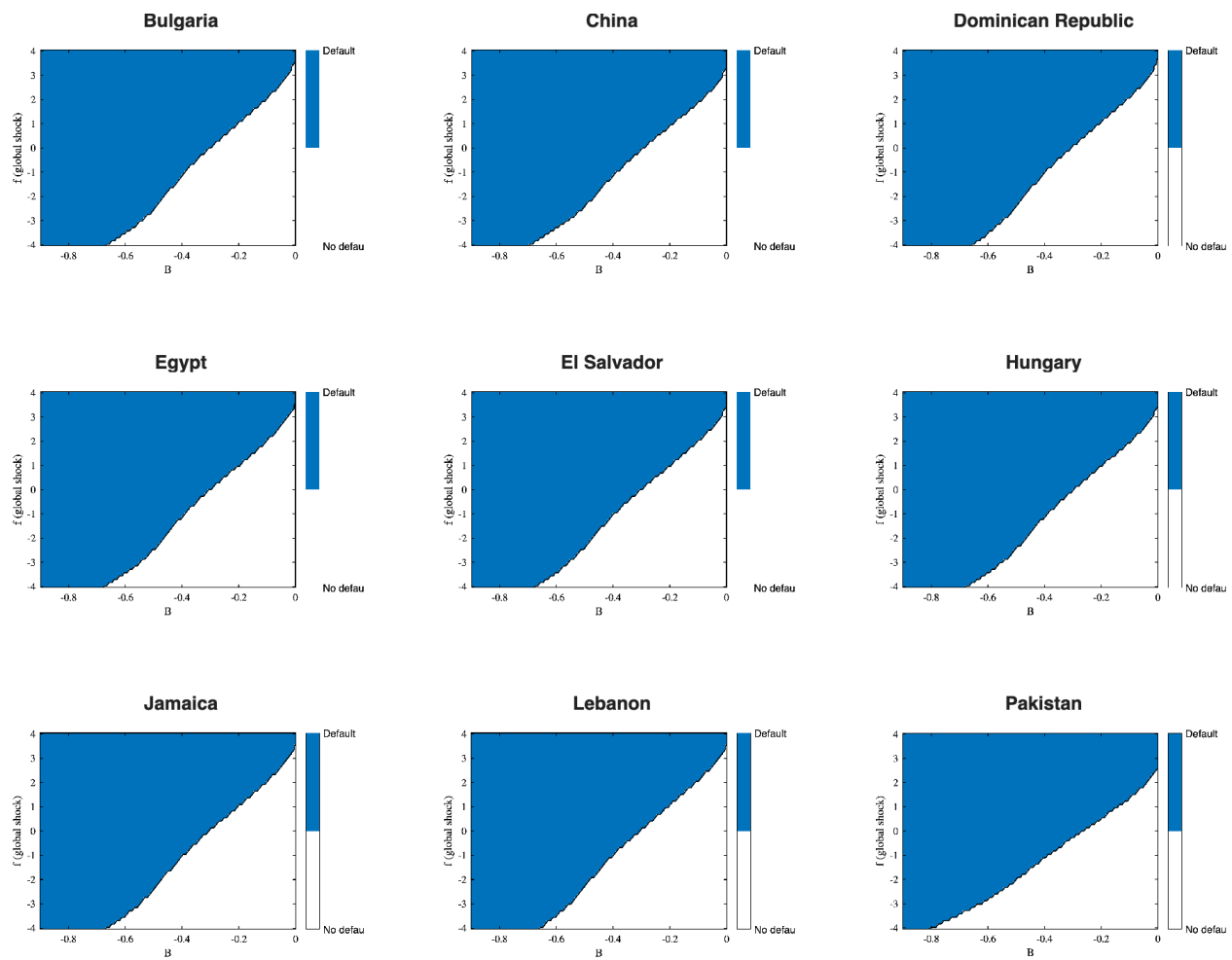
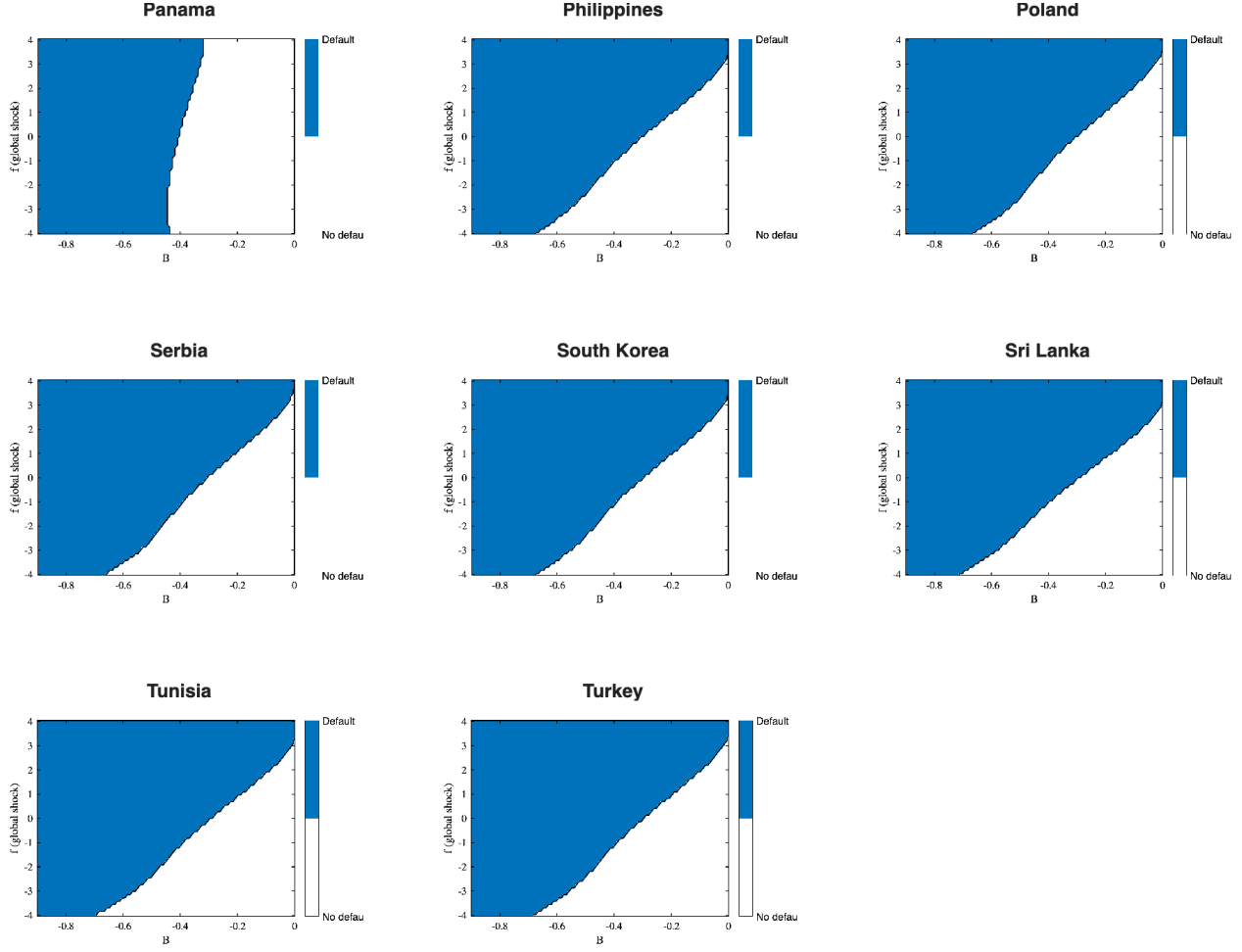


Figure 18: Default Sets – Importer-type Countries, Part 2.



F.3 Bond Price Schedules for the 37 Countries

Figures 19–22 report the equilibrium bond price schedules $q(f, b')$ for all countries. Each panel shows, for a given country, the price at which lenders are willing to purchase new debt b' as a function of the current global factor f .

Bond prices decline as b' becomes more negative, reflecting the higher default risk associated with greater borrowing. The dependence of prices on the global factor mirrors the structure of default incentives. In exporter-type countries, bond prices are increasing in f , so spreads decline when global commodity prices rise. In importer-type countries, bond prices are decreasing in f , implying that spreads increase with the global factor.

The sensitivity of bond prices to f varies across countries and increases with exposure to the global factor. Countries with larger absolute exposure display a wider separation between price schedules at low and high realizations of f , consistent with a stronger response of default

risk to global shocks.

Figure 19: Bond Price Schedules – Exporter-type Countries, Part 1.

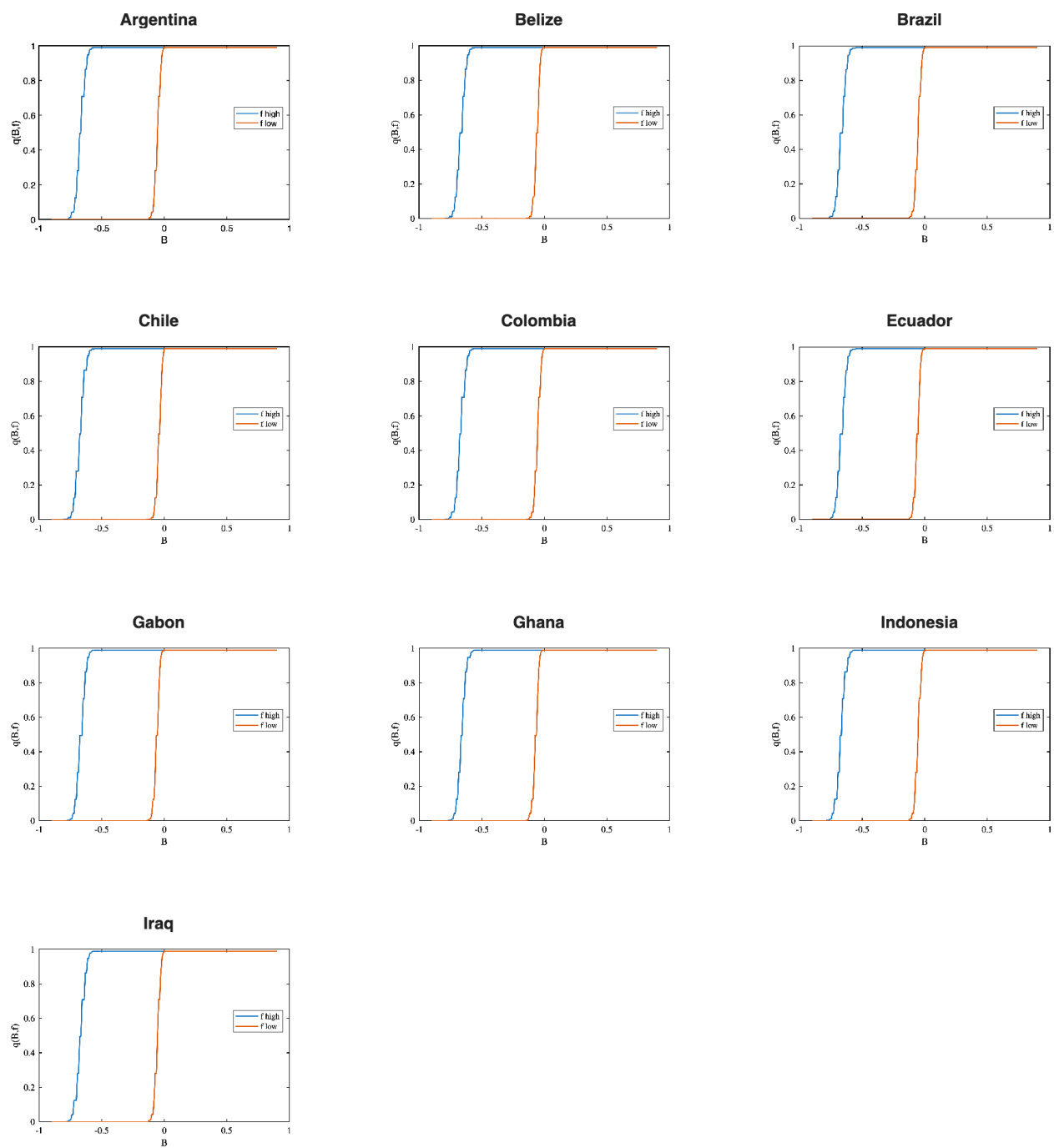


Figure 20: Bond Price Schedules – Exporter-type Countries, Part 2.

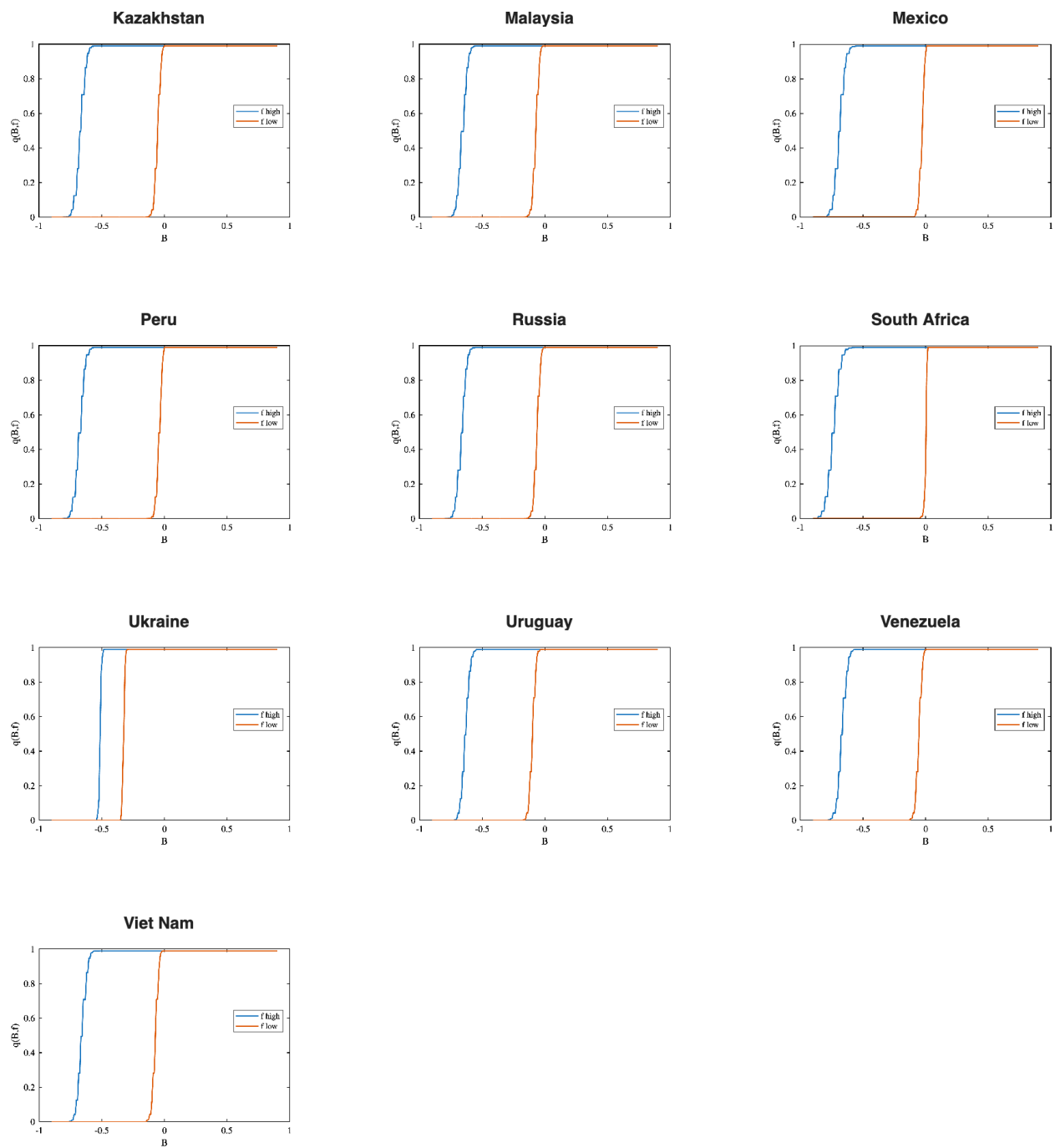


Figure 21: Bond Price Schedules – Importer-type Countries, Part 1.

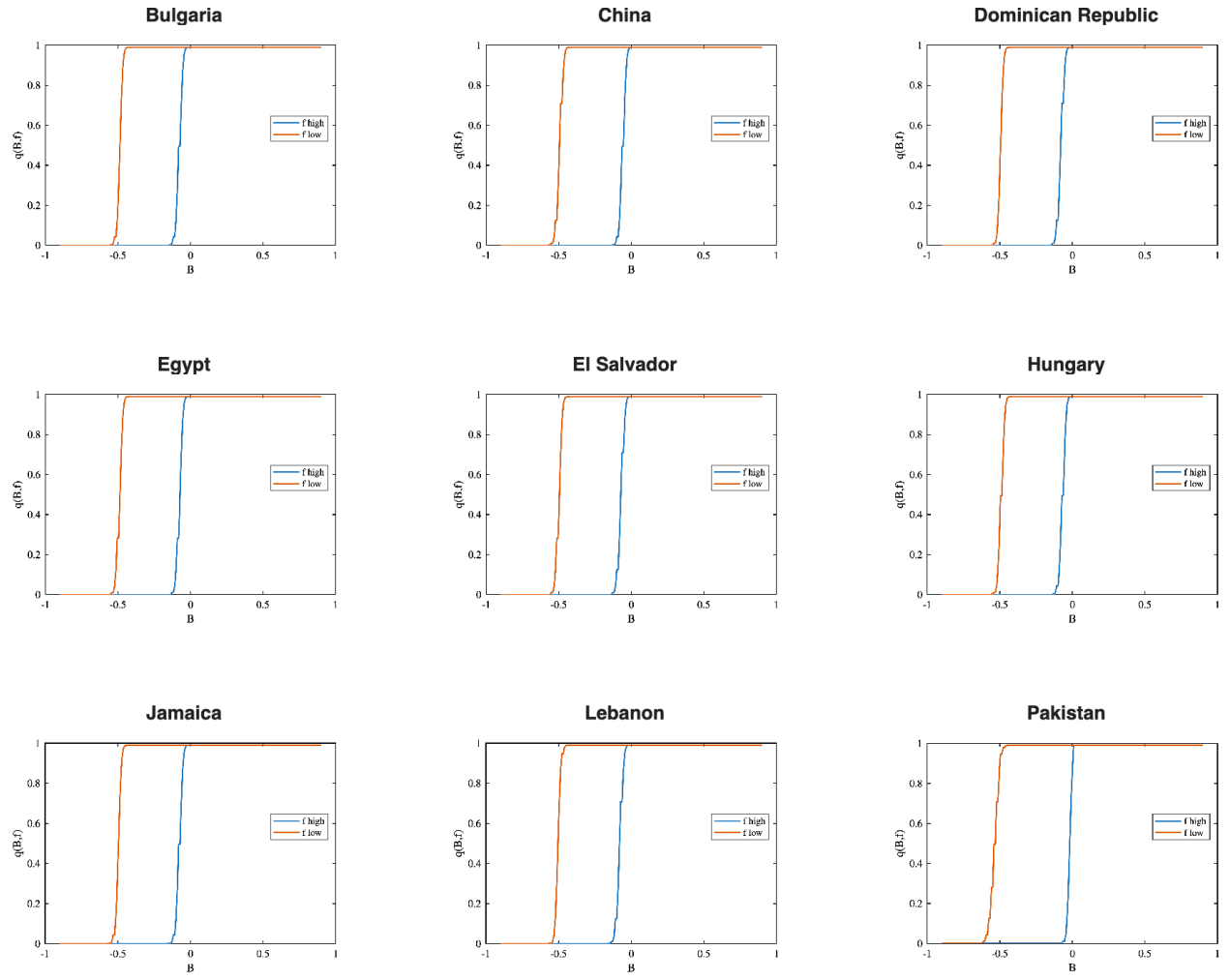
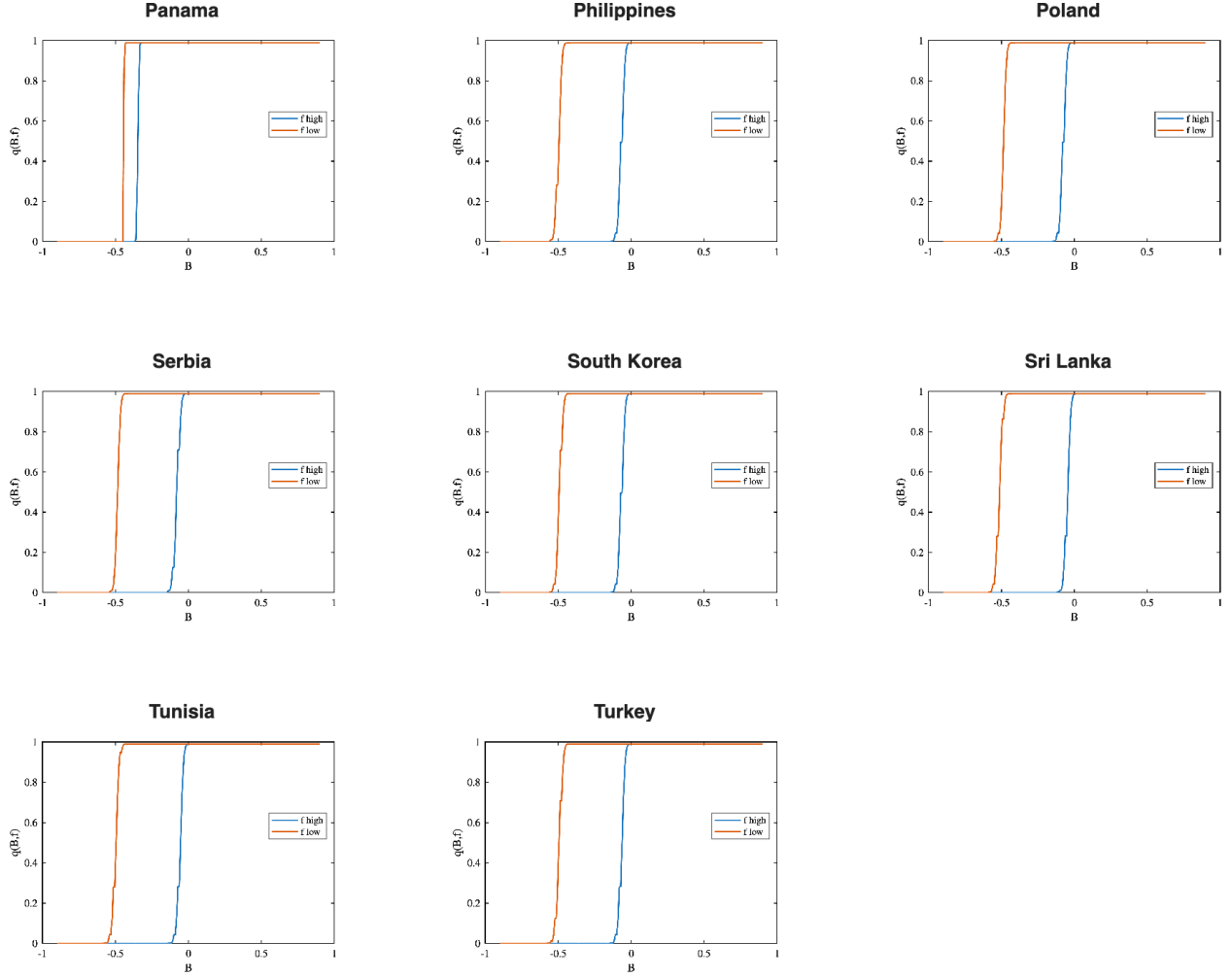


Figure 22: Bond Price Schedules – Importer-type Countries, Part 2.



F.4 Dynamics Around Default Events for the 37 Countries

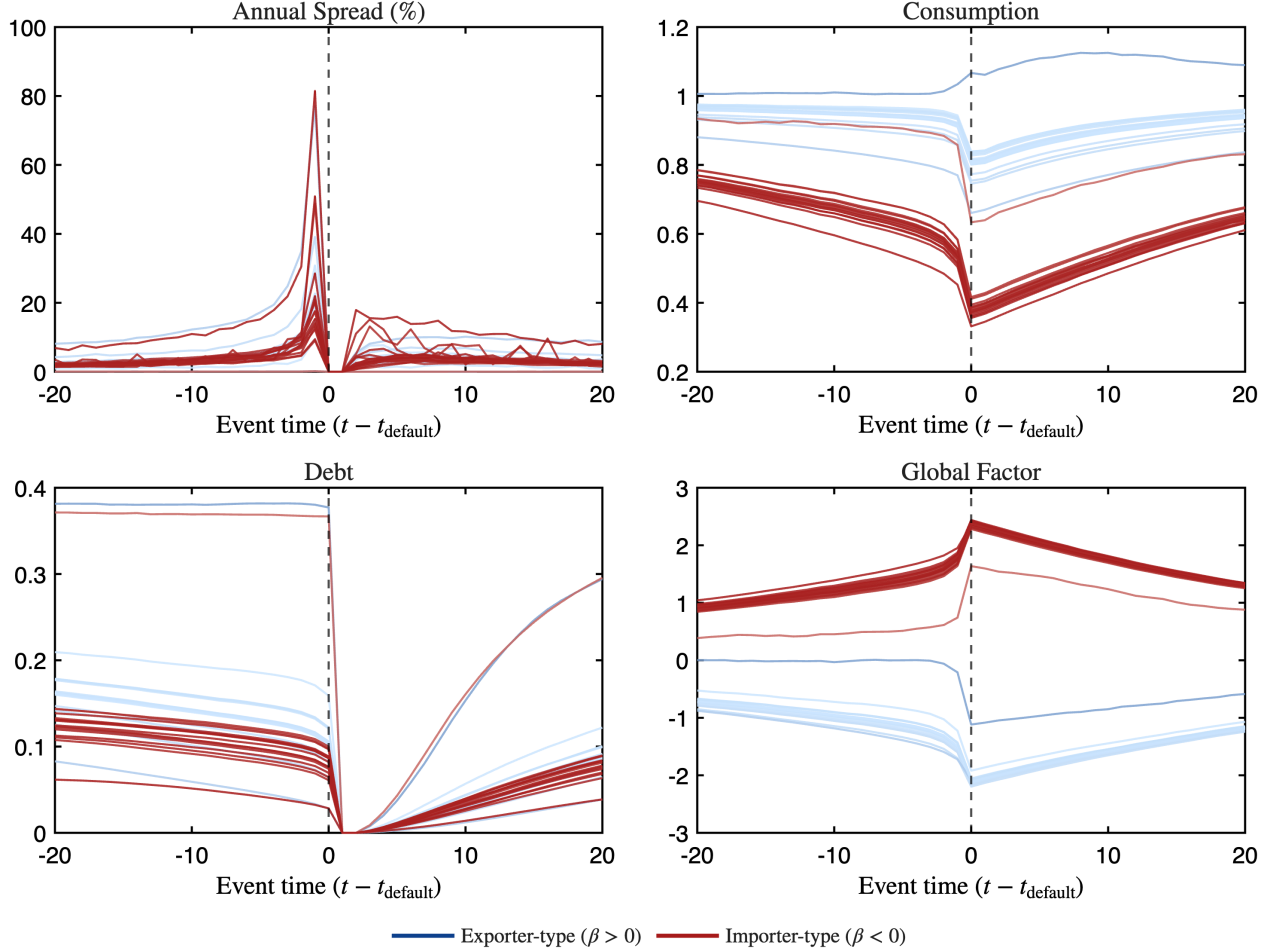
Figure 23 reports event-study dynamics for all countries, with blue shades for exporter-type calibrations and red shades for importer-type calibrations; color intensity reflects the magnitude of each country's exposure to the global factor. The vertical dashed line marks the default date.

The timing of default events differs systematically across groups. In exporter-type countries, defaults are preceded by negative realizations of the global factor: declining export prices weaken terms of trade, reduce real income, and raise spreads prior to default. In importer-type countries, defaults are preceded by positive realizations of the global factor: rising import prices increase the cost of the consumption basket, compress real income, and

widen spreads.

Across both groups, consumption reaches a trough around the time of default, while debt drops discontinuously at $\tau = 0$ as repayment is suspended. The magnitude of the pre-default increase in spreads rises with exposure to the global factor, reflecting the stronger response of default incentives to global shocks in more exposed economies.

Figure 23: Dynamics Around Default Events – All 37 Sample Countries.



Note: Each thin line shows the simulated average path of the indicated variable around default events for a given country. Colors denote wedge type (blue for exporter-type, red for importer-type), with intensity proportional to exposure to the global factor. Default events are based on 500 simulations of 10,000 periods per country.